Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

Yunjie Fan BRIANFAN@USC.EDU
Department of Mathematics
University of Southern California
Los Angeles, CA, USA

Matteo Sesia SESIA@MARSHALL.USC.EDU
Department of Data Sciences and Operations
Thomas Lord Department of Computer Science
University of Southern California
Los Angeles, CA, USA

Abstract

We propose a conformal prediction method for constructing tight simultaneous prediction intervals for multiple, potentially related, numerical outputs given a single input. This method can be combined with any multi-target regression model and guarantees finite-sample coverage. It is computationally efficient and yields informative prediction intervals even with limited data. The core idea is a novel *coordinate-wise* standardization procedure that makes residuals across output dimensions directly comparable, estimating suitable scaling parameters using the calibration data themselves. This does not require modeling of cross-output dependence nor auxiliary sample splitting. Implementing this idea requires overcoming technical challenges associated with transductive or full conformal prediction. Experiments on simulated and real data demonstrate this method can produce tighter prediction intervals than existing baselines while maintaining valid simultaneous coverage.

Keywords: interpretability, multivariate analysis, statistical inference, supervised learning, uncertainty quantification.

1 Introduction

1.1 Motivation and Preview of Contributions

We are interested in jointly quantifying uncertainty when predicting several, possibly dependent continuous outcomes from a shared set of input features. This problem is motivated by many applications where data-driven decisions depend on several unknown variables rather than on any one of them in isolation. For example, a financial institution may use available customer data to jointly forecast that customer's future credit-line utilization, monthly spending volume, and credit score. In this and many other contexts, decision quality hinges on the joint reliability of all predicted values, and an output that is incorrect in even one dimension could lead to adverse consequences.

Conformal prediction is a popular and broadly applicable framework for uncertainty quantification in predictive settings (Vovk et al., 2005; Angelopoulos and Bates, 2023). It derives its main strengths from the ability to augment the output of any machine-learning model with *prediction sets* equipped with finite-sample guarantees, without relying on strong assumptions on the data distribution. Its sole assumption is the availability of labeled data that are *exchangeable* with the test point of interest, a weaker requirement than the standard assumption of i.i.d. sampling. For a *univariate* setting, where the goal is to predict a single continuous-valued outcome, there now exist several well-established conformal prediction methods that produce valid, computationally fast, and easy-to-interpret prediction intervals (Lei et al., 2018; Romano et al., 2019; Sesia and Romano, 2021).

Extending conformal prediction to the *multivariate* setting is a non-trivial problem that has attracted considerable recent interest, spurring a variety of solutions (Dheur et al., 2025). The fundamental challenge is to aggregate a vector of generalized residuals—whose components may differ substantially in scale and variability—into a scalar-valued *non-conformity score* leading to *prediction sets* that should satisfy four key desiderata: (i) finite-sample coverage, (ii) informativeness or tightness, (iii) computational tractability, and (iv) interpretability. Interpretability can take different meanings depending on the application, and in some settings it is consistent with prediction sets having complex shapes. Nevertheless, in many practical scenarios, hyper-rectangular prediction sets—or, equivalently, jointly valid prediction intervals—provide the most natural and transparent way to convey uncertainty to practitioners. This paper focuses on such settings.

Existing methods achieve only a subset of these desiderata. The simplest approach to obtain hyper-rectangular prediction sets is to apply univariate conformal prediction to each output and then take the Cartesian product of the resulting intervals (Neeven and Smirnov, 2018); however, a conservative Bonferroni adjustment to the marginal miscoverage levels is needed to guarantee joint coverage; see Algorithm S1 in Appendix A. This strategy is very easy to implement, but the Bonferroni correction is often too conservative to be practically satisfactory, as it adopts a worst-case view of the dependence structure across output dimensions. By contrast, we aim to achieve valid coverage with joint prediction intervals that are as tight as possible; see Figure **??** for a sketch of this goal.

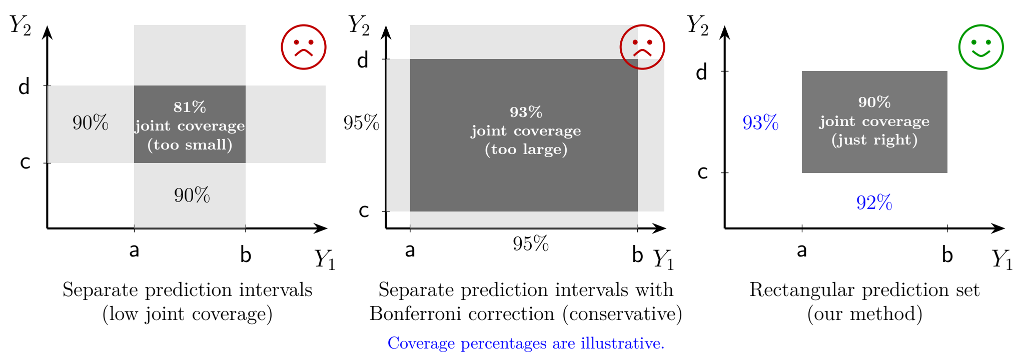

Figure 1: Two-dimensional toy example depicting graphically the main objective of this work: constructing *informative* and *interpretable* hyper-rectangular conformal prediction sets for multivariate outcomes. Left: univariate approaches can produce marginal prediction intervals for each output separately, which do not guarantee joint coverage. Center: a Bonferroni correction can easily restore valid joint coverage, at the cost of being overly conservative. Right: our goal is to efficiently construct hyper-rectangular prediction sets that are easy to interpret and retain valid coverage; doing so involves technical challenges.

A second class of existing methods reduce multivariate residuals to scalar-valued non-conformity scores, for example by applying an ℓ_2 or ℓ_∞ norm, then apply a univariate conformal prediction method to these scalar scores, and finally map the resulting prediction interval back into the original output space by inverting the dimension-reduction transformation; e.g., see Algorithm S2 in Appendix A. While computationally efficient and in principle able to produce hyper-rectangular prediction sets (e.g., using the ℓ_∞-norm), these approaches are dominated by the noisiest dimension, often leading to prediction sets that are unnecessarily wide along low-uncertainty directions.

More sophisticated approaches have been developed to directly minimize the volume of the resulting prediction sets; for example using optimal transport, learned embeddings, copula, or generative models to describe the dependence structure across residual dimensions. However, these methods aim to solve a problem that is different from that considered in this paper, since they are explicitly designed to produce non-rectangular prediction sets. Moreover, they often introduce substantial computational overhead and require additional independent training data to provide finite-sample guarantees.

In this paper we focus on targeting what we believe is the core difficulty in many applications: the mismatch in the *marginal* scale and variability of residuals across output dimensions. To address this challenge, we develop a method based on transductive coordinate-wise residual standardization. This restores comparability across coordinates and enables the construction of tight hyper-rectangular prediction sets that guarantee finite-sample coverage, without necessitating additional training data or prohibitive computational overhead. Remarkably, our method does *not* need to explicitly model residual dependencies across different dimensions, unlike existing approaches focusing on non-rectangular prediction sets.

Making this idea operational requires overcoming several hurdles, which we resolve through a sequence of increasingly practical steps. We begin by formulating an ideal oracle procedure that assumes access to the test point's true outcome vector and uses this information to standardize the multi-dimensional residuals in the calibration set without violating the exchangeability conditions that are necessary to guarantee finite-sample coverage. Although clearly impractical, this oracle is valuable to guide the design of our practical method. Inspired by the oracle procedure, we introduce two increasingly sophisticated data-driven approaches. The first provides a more conservative approximation of the oracle, whereas the second achieves greater statistical efficiency by building on top of the first. Through extensive numerical experiments, we demonstrate that this progression of ideas culminates in a practical method producing prediction sets satisfying all our desiderata.

1.2 Related Works

Extending conformal prediction to multivariate outcomes has attracted substantial attention in recent years, leading to the development of a wide range of alternative methods.

A prominent line of work seeks minimum-volume prediction sets with flexible geometries by modeling dependence across output dimensions. This includes copula-based approaches (Messoudi et al., 2021; Park and Cho, 2025; Sun and Yu, 2023), non-parametric density estimation methods (Sadinle et al., 2019; Izbicki et al., 2022; Wang et al., 2023; Dheur et al., 2025), parametric density estimation methods (Johnstone and Cox, 2021; Sampson and Chan, 2024; Braun et al., 2025a,b), optimal-transport techniques (Thurin et al., 2025; Klein et al., 2025), and flow-based generative models (Colombo, 2024; Lee et al., 2025; Fang et al., 2025). While these approaches can produce smaller prediction sets compared to our method, these often have complex shapes that are more difficult to interpret. Moreover, these approaches require additional model training on independent data to provide finite-sample guarantees, and many also involve substantial computational overhead.

To the best of our knowledge, the idea of directly addressing heterogeneity across output dimensions via marginal residual rescaling remains relatively under-explored. Baheri and Amiri Shahbazi (2025) pursue a related idea, but rely on a conservative Bonferroni-type correction. The approach of Sampson and Chan (2024) is more closely related to ours, but requires an additional independent dataset; avoiding this requirement is the main technical challenge addressed in our paper. As we show empirically, the method of Baheri and Amiri Shahbazi (2025) is often considerably less efficient than ours—particularly in small-sample regimes—resulting in less stable and less informative prediction sets.

Our solution can be viewed as implementing a special instance of transductive, or full, conformal prediction (Vovk et al., 2005; Vovk, 2013). In contrast to the more commonly used split conformal methods, full conformal prediction repeatedly incorporates the test point, augmented with different candidate labels, directly into a learning algorithm. Although this strategy is computationally prohibitive in general, especially for continuous-valued numerical outcomes and complex algorithms, it can be implemented efficiently in some cases by exploiting structural properties of the underlying predictive model. Accordingly, our paper relates to a broader literature on making full conformal inference computationally practical in special cases, including approaches tailored to sparse generalized linear models (Lei, 2019; Guha et al., 2023) and methods that leverage stability or other forms of structure in the predictive model (Ndiaye and Takeuchi, 2019; Abad et al., 2022; Martinez et al., 2023; Li, 2024). Unlike those works, which focus primarily on accelerating model refitting, we assume a fixed, pre-trained model and instead aim to improve efficiency by avoiding calibration data splitting while explicitly standardizing residuals across output dimensions. Both the problem setting and the proposed methodology are therefore novel.

More broadly, our approach is also related to recent conformal methods that reuse calibration data for different purposes, such as enabling early stopping to control overfitting when training deep learning models (Liang et al., 2023) and leveraging data-adaptive grouping to improve conditional coverage (Zhou and Sesia, 2024; Zhang and Candès, 2024). In contrast to these works, our focus is on addressing heterogeneity across targets in multivariate regression while obtaining interpretable, rectangular prediction sets.

1.3 Outline

This paper is organized as follows. Section 2 introduces notations, a formal problem statement, and necessary assumptions, along with a description of an ideal oracle approach. Section 3 presents our methodology, starting with a more conservative approach and then developing more efficient and computationally tractable implementations. Section 4 validates our method empirically and compares its performance to that of existing approaches using both simulated and real data. Section 5 discusses some limitations and possible directions for future research. The Appendices contain mathematical proofs not contained in the main text, additional algorithmic details, and further numerical results.

2 Problem Setup and Motivating Constructions

2.1 Notation

For any $n \in \mathbb{N}$, we denote $[n] := \{1, \ldots, n\}$. For any $m \in \mathbb{N}$, $\alpha \in (0,1)$, and $(v_1, \ldots, v_m) \in \mathbb{R}^m$, let $\widehat{Q}_{1-\alpha}(v_1, \ldots, v_m)$ denote the $\lceil (1-\alpha)(m+1) \rceil$-th smallest element of $\{v_1, \ldots, v_m, +\infty\}$.

2.2 Setup and Data Assumptions

We consider a multi-output (multivariate) regression setting where the goal is to predict a vector $\mathbf{Y} = (Y_1, \ldots, Y_d)$ of d real-valued outputs given input features $\mathbf{X} \in \mathcal{X}$. Assume that $(\mathbf{X}^1, \mathbf{Y}^1), \ldots, (\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$ are jointly exchangeable, or drawn i.i.d. from an arbitrary distribution $P_{X,Y}$. We observe the calibration data $\mathcal{D}_{\text{cal}} = \{(\mathbf{X}^i, \mathbf{Y}^i)\}_{i=1}^n$ and the test covariate $\mathbf{X}^{n+1}$, while the corresponding outcome $\mathbf{Y}^{n+1}$ remains unobserved. Using all of the observed data, the task is to construct a reasonably tight prediction set $\hat{C}(\mathbf{X}^{n+1}) \subset \mathbb{R}^d$ that contains the unknown output $\mathbf{Y}^{n+1}$ with high probability, guaranteeing *marginal coverage*:

$$\mathbb{P}\Big[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})\Big] \geq 1 - \alpha, \tag{1}$$

where $\alpha \in (0,1)$ is the user-specified miscoverage level. Note that both $\mathbf{Y}^{n+1}$ and $\hat{C}(\mathbf{X}^{n+1})$ are random in (1), as they depend on the new test point as well as on the calibration data $\mathcal{D}_{\text{cal}}$. The term "marginal coverage" reflects that the guarantee in (1) holds on average over the joint randomness of both calibration sample and the test point. Stronger notion of coverage, such as conditional coverage on which the probability in (1) is conditioned on some fixed values or ranges of the test point, are generally unattainable in finite samples without much more restrictive assumptions (Foygel Barber et al., 2021). There are some techniques to approximate the conditional coverage (Romano et al., 2020; Gibbs et al., 2025) that may be possible to combine with our approach and could be an interesting direction for future work, but it is beyond the scope of this paper.

For interpretability, we aim to enforce the prediction sets to be *hyper-rectangular*, or Cartesian product of univariate prediction intervals:

$$\hat{C}(\mathbf{X}^{n+1}) = \bigtimes_{j=1}^{d} \big[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})\big], \tag{2}$$

where $[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})]$ denotes the coordinate-wise interval for the jth output. Prediction sets in this form decompose naturally across dimensions, making them very easy to explain in the context of multi-output prediction. Consequently, the marginal coverage requirement stated in (1) is equivalent to

$$\mathbb{P}\left[Y_j^{n+1} \in [L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})] \text{ for all } j \in [d]\right] \geq 1 - \alpha.$$

Thus, although the coverage guarantee is marginal over the calibration and test data, it is simultaneous across all d output coordinates.

2.3 Coordinate-Wise Generalized Residuals

In addition to the data assumptions, we also assume access to a *pre-trained* joint predictive model $\hat{\mathbf{f}} = (\hat{f}_1, \ldots, \hat{f}_d)$, fitted on dataset independent of both the calibration set $\mathcal{D}_{\text{cal}}$ and the test point. This model is treated as a "black-box" and could be anything; the only requirement for $\hat{\mathbf{f}}$ is that it must provide us with a coordinate-wise *generalized residual* map $\mathbf{E} : \mathcal{X} \times \mathbb{R}^d \to \mathbb{R}^d$, which quantifies the model's prediction error along each output dimension.

Let $\mathbf{E}(\mathbf{X}, \mathbf{Y}) = \big(E_1(\mathbf{X}, Y_1), \dots, E_d(\mathbf{X}, Y_d)\big) \in \mathbb{R}^d$ denote a vector of coordinate-wise generalized residuals. A classical example for models $\hat{f}_j$ that estimate the conditional mean $\mathbb{E}[Y_j \mid \mathbf{X}]$ (Lei et al., 2018) is $E_j(\mathbf{X}, Y_j) = |Y_j - \hat{f}_j(\mathbf{X})|$. Alternatively, for models that estimate conditional quantiles, as in conformalized quantile regression (Romano et al., 2019), a natural choice is

$$E_j(\mathbf{X}, Y_j) = \max\big\{\hat{f}_j^{\mathrm{lo}}(\mathbf{X}) - Y_j,\ Y_j - \hat{f}_j^{\mathrm{hi}}(\mathbf{X})\big\},$$

where $\hat{f}_j = (\hat{f}_j^{\mathrm{lo}}, \hat{f}_j^{\mathrm{hi}})$ denote the estimated lower and upper conditional quantiles of $Y_j \mid \mathbf{X}$, for example at levels $\alpha/2$ and $1-\alpha/2$. Intuitively, this choice of E_j measures the signed distance of Y_j to the nearest boundary of the interval $[\hat{f}_j^{\mathrm{lo}}(\mathbf{X}), \hat{f}_j^{\mathrm{hi}}(\mathbf{X})]$, where positive values indicate that Y_j falls outside the interval, while negative values indicate that it lies inside.

Applying the generalized residual map to the observed calibration data gives $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i) = (E_1^i, \dots, E_d^i)$ for all $i \in [n]$. We reserve $\mathbf{E}^{n+1} = \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$ for the unobserved residual vector of the test point.

Throughout this paper, we would like to assume that the residuals are non-negative. For the types of residuals with possibly negative coordinates, such as the quantile-based residuals reviewed above, we can always make them positive by truncating at zero, with the cost of potentially more conservative conformal prediction sets (Romano et al., 2019).

We also impose two additional regularity conditions. First, each residual coordinate is assumed to have finite mean and finite, strictly positive variance at the population level. These moment conditions are mild and automatically satisfied by bounded and nondegenerate residuals. Second, within each coordinate, the calibration and test residuals are assumed to be almost surely distinct, which is a standard assumptionin conformal inference. When ties arise, including the ties at zero that may be introduced by truncation, they can be handled through standard randomized tie-breaking or by adding independent continuous jitter to the residuals. These conditions make the theoretical development cleaner, while our numerical experiments indicate that the proposed method remains robust when they are not exactly satisfied. We collect all of the standing conditions below.

Assumption 1 *For every $j \in [d]$, the distribution of the residual $E_j(\mathbf{X}, Y_j)$ satisfies $E_j(\mathbf{X}, Y_j) \geq 0$ almost surely, $\mathbb{E}[E_j(\mathbf{X}, Y_j)] < \infty$, and $0 < \mathrm{Var}[E_j(\mathbf{X}, Y_j)] < \infty$. Moreover, for every $j \in [d]$, the residuals $E_j^1, \dots, E_j^{n+1}$ are almost surely pairwise distinct:*

$$\mathbb{P}\left(E_j^i \neq E_j^k \text{ for all } 1 \leq i < k \leq n+1\right) = 1.$$

2.4 Conformal Calibration with Marginal Coverage

After evaluating the residuals $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i)$ for all calibration points indexed by $i \in [n]$, a prediction set for $\mathbf{Y}^{n+1}$ based on $\mathbf{X}^{n+1}$ can be assembled by including all possible values of $\mathbf{y} \in \mathbb{R}^d$ whose residuals $\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})$ are sufficiently small:

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq W_j,\ \text{for all } j \in [d]\right\}, \tag{3}$$

where $W_1, \dots, W_d$ are suitable (data-dependent) thresholds chosen to satisfy

$$\mathbb{P}\left[0 \leq E_j(\mathbf{X}^{n+1}, Y_j^{n+1}) \leq W_j \text{ for all } j \in [d]\right] \geq 1 - \alpha. \tag{4}$$

This will ensure marginal coverage of $\mathbf{Y}^{n+1}$ at level $1-\alpha$, as defined in (1). For standard choice of $\mathbf{E}$, such as the absolute residuals, the resulting prediction set $\hat{C}(\mathbf{X}^{n+1})$ is hyper-rectangular, as in (2), with $L_j(\mathbf{X}^{n+1}) = \hat{f}_j(\mathbf{X}^{n+1}) - W_j$ and $U_j(\mathbf{X}^{n+1}) = \hat{f}_j(\mathbf{X}^{n+1}) + W_j$ for all $j \in [d]$. Similar for the quantile-based residuals described above.

A naive way to achieve (4) is to set $W_j = \hat{Q}_{1-\alpha/d}(E_j^1, \dots, E_j^n)$, which is equivalent to applying univariate conformal prediction separately across all d outcome dimensions, with a Bonferroni correction (α/d replacing α). This typically leads to very conservative prediction sets because it takes an overly pessimistic worst-case view of possible dependencies of the prediction tasks across outcome dimensions.

In this paper, we pursue a more efficient approach, started by computing induced *scalar* non-conformity scores $S^i = \Phi(\mathbf{E}^i) \in \mathbb{R}$ for all $i \in [n]$ using a scalarization map $\Phi : \mathbb{R}_+^d \to \mathbb{R}$, whose form will be discussed below. Then, the prediction set is constructed as:

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{\mathbf{y} \in \mathbb{R}^d : \Phi(\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})) \leq \hat{Q}_{1-\alpha}(S^1, \dots, S^n)\right\}. \tag{5}$$

If Φ is, in addition, an invertible map, then we can recover the form of (3) from (5) by setting $W_j = \Phi_j^{-1}\left(\hat{Q}_{1-\alpha}(S^1, \dots, S^n)\right)$. We focus on the case where Φ is invertible in this paper, but the coverage guarantee is generally valid for other type of Φ as demonstrated in the following theorem.

Theorem 1 (Generalized Marginal Coverage) *Assume* $(\mathbf{E}^1, \dots, \mathbf{E}^{n+1})$ *are exchangeable, with* $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i)$ *for all* $i \in [n+1]$. *Let* $\Phi : \mathbb{R}_+^d \to \mathbb{R}$ *be any* fixed *function, independent of the data. Then,* $\hat{C}(\mathbf{X}^{n+1})$ *defined in* (5) *satisfies* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \geq 1-\alpha$. *Moreover, if the scores* $(S^1, \dots, S^{n+1})$ *given by* $S^i = \Phi(\mathbf{E}^i)$ *are almost-surely distinct,* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \leq 1 - \alpha + \frac{1}{n+1}$. *The same results hold with* Φ *random, if* $(S^1, \dots, S^{n+1})$ *are exchangeable conditional on it.*

If the induced scalar scores $S^1, \dots, S^{n+1}$ are almost surely distinct, Theorem 1 guarantees the prediction sets defined in (5) achieve marginal coverage *tightly* from above, as long as the sample size n is not too small. A limitation of Theorem 1, however, is that it cannot tell us whether the coverage is equally tight across different outcome dimensions. In fact, depending on the data distribution, the fitted model, and the choice of Φ, it could happen that $\hat{C}(\mathbf{X}^{n+1})$ achieves the desired marginal coverage while being much wider than necessary along certain directions, thereby providing uncertainty estimates that are potentially much less informative than ideal. Moreover, depending on Φ, the prediction set $\hat{C}(\mathbf{X}^{n+1})$ may have a difficult-to-interpret shape. Several recent works propose residual transformations based on ℓ_p-norms, i.e., $\Phi(\cdot) = \|\cdot\|_p$ for different $1 \leq p \leq \infty$ (Johnstone and Ndiaye, 2022). Although conceptually and computationally simple, these can be heavily affected by the noisiest output dimension. A single high-variance dimension may dominate the scalar score and inflate the resulting prediction set uniformly across all coordinates. To obtain tighter prediction sets, recent works have proposed learning a suitable transformation of the residuals prior to applying a norm-based dimension reduction, for example by using optimal transport (Thurin et al., 2025; Klein et al., 2025). However, these approaches require additional training data as well as expensive computations, and they typically lead to prediction sets with irregular shapes.

In this paper, we propose a conceptually simpler solution that consists of *standardizing* the residuals prior to reducing their dimension by taking the maximum across coordinates, similar to the ℓ_∞ norm. This leads to rectangular prediction sets with balanced coordinatewise coverage across all output dimensions, without requiring additional training data or expensive computations. Despite the apparent simplicity of this idea, however, achieving all desiderata while maintaining guaranteed finite-sample coverage involves significant technical challenges, which we overcome as gradually explained below.

2.5 A Population-Standardized Oracle

To motivate our method, we first consider an oracle construction that has access to the coordinate-wise population means and standard deviations of the residuals. Although unavailable in practice, this construction illustrates how coordinate-wise standardization can mitigate heterogeneity in residual scales while preserving validity and rectangular interpretability.

Consider prediction sets of the form in (5), obtained by transforming the vector-valued residuals $\mathbf{E}^i$'s using $\Phi : \mathbb{R}_+^d \to \mathbb{R}$:

$$\Phi(\mathbf{t}; \boldsymbol{\beta}, \boldsymbol{\theta}) := \max_{1 \leq j \leq d} \frac{t_j - \beta_j}{\theta_j}, \tag{6}$$

where $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_d)^\top$ and $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_d)^\top$ are vectors of location and scale parameters, respectively. This transformation is invertible, with coordinate-wise inverse $\Phi_j^{-1}(c; \boldsymbol{\beta}, \boldsymbol{\theta}) = c \cdot \theta_j + \beta_j$ for any $c \in \mathbb{R}$. Given these two parameters, we define the $1 - \alpha$ level prediction set of $\mathbf{X}^{n+1}$ by

$$\begin{gathered} C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta}) = \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha}^{\boldsymbol{\beta}, \boldsymbol{\theta}} \cdot \theta_j + \beta_j \text{ for all } j \in [d] \right\}, \\ \text{where} \quad \hat{Q}_{1-\alpha}^{\boldsymbol{\beta}, \boldsymbol{\theta}} := \hat{Q}_{1-\alpha}(S_{\boldsymbol{\beta}, \boldsymbol{\theta}}^1, \ldots, S_{\boldsymbol{\beta}, \boldsymbol{\theta}}^n) \quad \text{and} \quad S_{\boldsymbol{\beta}, \boldsymbol{\theta}}^i = \Phi(\mathbf{E}^i; \boldsymbol{\beta}, \boldsymbol{\theta}). \end{gathered} \tag{7}$$

This set is hyper-rectangular and aligns the form of (3) with coordinate-wise thresholds $W_j = \Phi_j^{-1}(\hat{Q}_{1-\alpha}^{\boldsymbol{\beta}, \boldsymbol{\theta}}; \boldsymbol{\beta}, \boldsymbol{\theta})$. Theorem 1 guarantees that $C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta})$ satisfies the marginal coverage requirement in (1) for any fixed choice of $(\boldsymbol{\beta}, \boldsymbol{\theta})$.

To address the possible heterogeneity in scale and variability across different dimensions, the most natural option of $(\boldsymbol{\beta}, \boldsymbol{\theta})$ are the population mean and standard deviation of the residual vector $\mathbf{E}(\mathbf{X}, \mathbf{Y})$, which we denote by $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$:

$$\boldsymbol{\mu}_j^* = \mathbb{E}[E_j(\mathbf{X}, Y_j)] \geq 0, \qquad \boldsymbol{\sigma}_j^* = \sqrt{\mathrm{Var}[E_j(\mathbf{X}, Y_j)]} > 0, \qquad \forall j \in [d]. \tag{8}$$

Then by replacing $(\boldsymbol{\beta}, \boldsymbol{\theta})$ in (7) with the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$, we obtain a population-standardized oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$.

Although practically unfeasible, this oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ satisfies all four of our desiderata: (i) it guarantees finite-sample marginal coverage, as per Theorem 1;

(ii) it tends to have balanced coordinatewise coverage across all output dimensions, because it operates with residuals whose components are all on the same scale; (iii) it is fast to compute, at least for standard choices of the residual function $\mathbf{E}$; (iv) it is an easy-to-interpret hyper-rectangle, at least for standard choices of $\mathbf{E}$.

In the following, we will describe practical approximations of these oracle prediction sets, starting from intuitive but not fully satisfactory solutions before presenting our method.

2.6 Simple but Unsatisfactory Practical Approaches

An intuitive *plug-in* approximation of the ideal oracle is obtained by replacing the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ with their empirical counterparts $(\hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$, evaluated on the observed calibration data, and then plugging in $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$ into the prediction set formula (7): $\hat{C}^{\text{pi}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$. This approach, outlined by Algorithm S3 in Appendix A, tends to perform well in large-samples (as shown empirically in Appendix C) but lacks finite-sample guarantees. In fact, using a data-dependent transformation Φ, which breaks the exchangeability between calibration and test scores, violates the key assumption of Theorem 1.

An alternative solution that is also intuitive yet unsatisfactory consists of combining the above plug-in approach with an additional data splitting step that allows one to recover finite-sample guarantees: the location and scale parameter estimates for the transformation function Φ, denoted here as $\hat{\boldsymbol{\mu}}^{\mathbf{ds}}$ and $\hat{\boldsymbol{\sigma}}^{\mathbf{ds}}$, are estimated empirically using only a random subset of $n_1 < n$ observations of the observed calibration data. The remaining $n_2 = n - n_1$ data points are used to evaluate the quantile $\hat{Q}_{1-\alpha}^{\hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}}}$ and construct $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}})$. This approach is outlined by Algorithm S4 in Appendix A. While it satisfies the key assumption of Theorem 1, it makes an inefficient use of the data. Since it reduces the effective number of calibration samples roughly by half, it tends to produce more unstable and often wider-than-necessary prediction sets, especially in small-sample regimes.

The main contribution of this paper is therefore to overcome this inefficiency. We develop a *transductive* conformal prediction method that uses a similar standardization idea but avoids wasteful data splitting by reusing the entire calibration data to estimate the scaling parameters, all while retaining finite-sample guarantees and manageable computational costs. Achieving this requires resolving several technical challenges, as explained step by step below.

3 Methodology

3.1 A Transductive Rescaling Oracle and High-Level Method Blueprint

According to Theorem 1, prediction sets of the form (5) achieve finite-sample marginal coverage if the dimension-reduction map Φ is independent of the calibration data, as assumed in the previous section, or if $(\Phi(\mathbf{E}^1), \ldots, \Phi(\mathbf{E}^{n+1}))$ can remain *exchangeable conditional on* Φ.

This latter requirement, which is strictly weaker than independence, is the foundation of the method developed in this section. We begin by showing how this conditional-exchangeability condition naturally leads to a different *transductive* oracle, which is much more closely aligned with the practical method introduced below.

Consider an imaginary oracle that knows the unordered values in $\{\mathbf{E}^1, \dots, \mathbf{E}^{n+1}\}$. We can estimate the population location and scale parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ defined in (8) as follows:

$$\hat{\boldsymbol{\mu}}^{\text{oracle}} = (\hat{\mu}_1^{\text{oracle}}, \dots, \hat{\mu}_d^{\text{oracle}})^\top, \quad \hat{\boldsymbol{\sigma}}^{\text{oracle}} = (\hat{\sigma}_1^{\text{oracle}}, \dots, \hat{\sigma}_d^{\text{oracle}})^\top,$$
$$\hat{\mu}_j^{\text{oracle}} = \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i, \quad \hat{\sigma}_j^{\text{oracle}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \hat{\mu}_j^{\text{oracle}}\right)^2}. \tag{9}$$

Since these statistics are *symmetric* in all $n+1$ data points and thus invariant to their ordering, conditioning on the random function $\Phi^{\text{oracle}}(t) := \Phi(t; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ maintains the exchangeability of $(S^1_{\text{oracle}}, \dots, S^{n+1}_{\text{oracle}})$ where $S^i_{\text{oracle}} := \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$. Therefore, Theorem 1 implies that the prediction set $C^{\text{oracle}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$, obtained by substituting $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ into (7), satisfies the marginal coverage requirement in (1). Of course, this oracle is still impractical because both Φ^{oracle} and $\hat{Q}^{\text{oracle}}_{1-\alpha} := \hat{Q}_{1-\alpha}(S^1_{\text{oracle}}, \dots, S^n_{\text{oracle}})$ depend on the unknown test outcome $\mathbf{Y}^{n+1}$ through $\mathbf{E}^{n+1}$.

We will now translate this oracle into a practical method producing a (slightly) larger prediction set that depends only on the observed data. First, we will find suitable *monotonically nondecreasing link functions* $\omega_1, \dots, \omega_d : \mathbb{R} \to \mathbb{R}_+$ satisfying

$$S^{n+1}_{\text{oracle}} \le c \iff 0 \le E_j^{n+1} \le \omega_j(c) \text{ for all } j \in [d], \tag{10}$$

for every $c \in \mathbb{R}$. Using these link functions, we can define an estimated $1-\alpha$ level prediction set of $\mathbf{X}^{n+1}$ by

$$\hat{C}(\mathbf{X}^{n+1}; c) = \left\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \omega_j(c) \text{ for all } j \in [d]\right\}. \tag{11}$$

Substituting $c = \hat{Q}^{\text{oracle}}_{1-\alpha}$ into (10) and (11) gives the equivalence relation

$$\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1}) \iff \mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}). \tag{12}$$

Although $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha})$ is still impractical, it is closer to being computable. The last remaining step is to replace $\hat{Q}^{\text{oracle}}_{1-\alpha}$ with a conservative data-driven estimate, which we denote here as $\hat{Q}^{\text{prac}}_{1-\alpha} \ge \hat{Q}^{\text{oracle}}_{1-\alpha}$. Because $\omega_1(c), \dots, \omega_d(c)$ are non-decreasing in c, doing that will give a practical prediction set

$$\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{prac}}_{1-\alpha}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}), \tag{13}$$

hence inheriting the coverage lower bound from Theorem 1. Technical difficulties, which we address below, arise when one tries to keep this approximation tight and computationally efficient.

3.2 Explicit Form of the Link Functions

The link functions $\omega_1, \dots, \omega_d$ defined implicitly in (10) can be derived by solving the inequalities in (10) for E_j^{n+1} in terms of c, through the observed statistics $\hat{\mu}_j^{\text{obs}}$ and $\hat{\sigma}_j^{\text{obs}}$ defined as

$$\hat{\mu}_j^{\text{obs}} = \frac{1}{n}\sum_{i=1}^{n} E_j^i, \quad \hat{\sigma}_j^{\text{obs}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(E_j^i - \hat{\mu}_j^{\text{obs}}\right)^2}, \quad \forall j \in [d]. \tag{14}$$

The explicit form of $\omega_j(c)$ is given in the following lemma.

Lemma 2 *Under Assumption 1, suitable link functions $\omega_1, \dots, \omega_d$ satisfying (10) are:*

$$\omega_j(c) = \begin{cases} 0, & \text{if } c \leq -\frac{n}{\sqrt{n+1}}, \\ \max\left\{0,\ \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\dfrac{(n+1)^2}{n^2-(n+1)c^2}}\right\}, & \text{if } -\dfrac{n}{\sqrt{n+1}} < c < 0, \\ \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\dfrac{(n+1)^2}{n^2-(n+1)c^2}}, & \text{if } 0 \leq c < \dfrac{n}{\sqrt{n+1}}, \\ \infty, & \text{if } c \geq \dfrac{n}{\sqrt{n+1}}, \end{cases} \qquad \forall j \in [d].$$

Replacing this expression for the link functions in (11) and substituting $c = \hat{Q}_{1-\alpha}^{\text{oracle}}$ give an upper bound $\omega_j(\hat{Q}_{1-\alpha}^{\text{oracle}})$ for $E_j(\mathbf{X}^{n+1}, y_j)$ that is finite (and hence informative) if and only if $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$. Fortunately, $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely as long as the sample size is not too small.

Lemma 3 *Under Assumption 1, if $n \geq 1/\alpha - 1$, then $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely.*

Armed with Lemma 2 and Lemma 3, what remains to be done is to find a tight upper bound $\hat{Q}_{1-\alpha}^{\text{prac}}$ for $\hat{Q}_{1-\alpha}^{\text{oracle}}$, as previewed in (13). This is the main challenge and focus of the next sections.

3.3 A Conservative Global-Worst-Case Approach

We first introduce a (highly) conservative practical upper bound $\hat{Q}_{1-\alpha}^{\text{gwc}}$, obtained by taking the worst-case upper bound of $\hat{Q}_{1-\alpha}^{\text{oracle}}$ over all possible values of $\mathbf{E}^{n+1} \in \mathbb{R}_+^d$. We refer to the resulting *global-worst-case* (GWC) prediction set as $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{gwc}})$. As we show in the next section, this conservative prediction set can in fact be sharpened considerably, without sacrificing marginal coverage, albeit with some additional methodological effort.

Let $\mathbf{z} \in \mathbb{R}_+^d$ denote a placeholder for the test residual $\mathbf{E}^{n+1}$. Define the plug-in scaling parameters $\hat{\boldsymbol{\mu}}(\mathbf{z}) := (\hat{\mu}_1(z_1), \dots, \hat{\mu}_d(z_d))$ and $\hat{\boldsymbol{\sigma}}(\mathbf{z}) := (\hat{\sigma}_1(z_1), \dots, \hat{\sigma}_d(z_d))$ as if $\mathbf{E}^{n+1}$ in (9) were replaced by $\mathbf{z}$:

$$\hat{\mu}_j(z_j) = \frac{\sum_{i=1}^n E_j^i + z_j}{n+1}, \quad \hat{\sigma}_j(z_j) = \sqrt{\frac{\sum_{i=1}^n \left(E_j^i - \hat{\mu}_j(z_j)\right)^2 + \left(z_j - \hat{\mu}_j(z_j)\right)^2}{n}}. \tag{15}$$

With this notation, we have $\hat{\boldsymbol{\mu}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\mu}}^{\text{oracle}}$ and $\hat{\boldsymbol{\sigma}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\sigma}}^{\text{oracle}}$. Define then the GWC transformation $\hat{\Phi}^{\text{gwc}} : \mathbb{R}_+^d \to \mathbb{R}$ as

$$\hat{\Phi}^{\text{gwc}}(\mathbf{t}) := \max_{1 \leq j \leq d} \sup_{z_j \geq 0} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \geq \Phi^{\text{oracle}}(\mathbf{t}), \qquad \forall \mathbf{t} \in \mathbb{R}_+^d. \tag{16}$$

Intuitively, for each coordinate $j \in [d]$ and $\mathbf{t} \in \mathbb{R}_+^d$, we consider the most unfavorable (largest) possible value of the standardized ratio as a function of $E_j^{n+1} = z_j \geq 0$. This ensures that $\hat{\Phi}^{\text{gwc}} \geq \Phi^{\text{oracle}}$ uniformly. Conveniently, $\hat{\Phi}^{\text{gwc}}$ admits a closed-form expression.

Lemma 4 *The GWC transformation* $\hat{\Phi}^{\text{gwc}}$ *defined in* (16) *can be written as:*

$$\hat{\Phi}^{\text{gwc}}(\mathbf{t}) = \max_{1\leq j\leq d} \max\left\{ \frac{t_j - \hat{\mu}_j(0)}{\hat{\sigma}_j(0)}, \frac{t_j - \hat{\mu}_j(z_j^*)}{\hat{\sigma}_j(z_j^*)}, -\frac{1}{\sqrt{n+1}} \right\}.$$

where $z_j^* = \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}$ *if* $\hat{\mu}_j^{\text{obs}} \geq \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}$ *and* $z_j^* = 0$ *otherwise.*

Using Lemma 4, we compute the scalar non-conformity scores $S_{\text{gwc}}^i := \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ for $i \in [n]$, and define the corresponding conformal quantile $\hat{Q}_{1-\alpha}^{\text{gwc}} := \hat{Q}_{1-\alpha}(S_{\text{gwc}}^1, \ldots, S_{\text{gwc}}^n)$. Because $\hat{\Phi}^{\text{gwc}} \geq \Phi^{\text{oracle}}$, it follows that $\hat{Q}_{1-\alpha}^{\text{gwc}} \geq \hat{Q}_{1-\alpha}^{\text{oracle}}$ almost surely. Substituting $c = \hat{Q}_{1-\alpha}^{\text{gwc}}$ in (11), we obtain a practical prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{gwc}})$ that is guaranteed to cover the unknown test outcome $\mathbf{Y}^{n+1}$ with probability at least $1-\alpha$. Algorithm S5 in Appendix A summarizes this procedure.

Theorem 5 *Assume* $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ *are exchangeable and Assumption 1 holds. Then,* $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ *constructed by Algorithm S5 satisfies:* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})] \geq 1-\alpha$.

Proof From (11), it is easy to see that $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ almost surely because $\hat{Q}_{1-\alpha}^{\text{gwc}} \geq \hat{Q}_{1-\alpha}^{\text{oracle}}$ and the link functions $\omega_1, \ldots, \omega_d$ outlined in Lemma 2 are monotonically non-decreasing. Then Theorem 1 and the dual relation in (12) together imply that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})] = \mathbb{P}[\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1})] \geq 1-\alpha$. ■

We refer to Appendix A.3.1 for details on how to construct $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ in practice at cost $\mathcal{O}(dn)$. While computationally efficient, however, $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ is *statistically* inefficient as it tends to be substantially larger than the estimated oracle prediction set $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$, which we would ideally like to approximate tightly. In the next section, we therefore extend this approach and develop a tighter *local worst-case* (LWC) prediction set $\hat{C}_{\mathcal{P}}^{\text{lwc}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ that approximates the oracle $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ more closely.

3.4 A Tighter Local-Worst-Case Method

The construction of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ described in the previous section relies on a *global* upper bound $\hat{Q}_{1-\alpha}^{\text{gwc}}$ for the ideal quantile $\hat{Q}_{1-\alpha}^{\text{oracle}}$, designed to hold uniformly over all possible values of the unobserved test residual $\mathbf{E}^{n+1} \in \mathbb{R}_+^d$. However, the resulting prediction sets are often overly conservative, as the GWC procedure treats all realizations of $\mathbf{E}^{n+1}$ equally, even if some choices correspond to $\mathbf{y}$'s that are extremely unlikely to satisfy the condition stated in $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ and could therefore be safely discarded.

This motivates the development of a more statistically efficient approach that utilizes a collection of *local* prediction sets, each tailored to a distinct working hypothesis that the true outcome $\mathbf{Y}^{n+1}$ belongs to a specific region of the outcome space. We will first present the high-level overview of this method and then detail the implementation in three steps: (1) we introduce an intuitive, data-driven partition of the outcome space guided by the order statistics of the calibration residuals; (2) we derive a tractable explicit characterization of the local prediction sets; (3) we develop a computationally efficient algorithm for constructing a tight enclosing rectangular prediction set in practice, while avoiding explicit enumeration of an exponential number of local prediction sets.

3.4.1 Method Overview

For any subset $\mathcal{R} \subseteq \mathbb{R}^d$, we consider the working hypothesis that $\mathbf{Y}^{n+1} \in \mathcal{R}$. Let $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})$ denote an upper bound for $\hat{Q}^{\text{oracle}}_{1-\alpha}$ tailored to this hypothesis, which satisfies

$$\mathbf{Y}^{n+1} \in \mathcal{R} \quad \Longrightarrow \quad \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R}) \geq \hat{Q}^{\text{oracle}}_{1-\alpha}. \tag{17}$$

Substituting $c = \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})$ in (11), we get a local prediction set

$$\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})) \cap \mathcal{R} \tag{18}$$

that satisfies

$$\mathbf{Y}^{n+1} \in \mathcal{R} \implies \mathcal{R} \supseteq \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \cap \mathcal{R}. \tag{19}$$

It is worth noting that the containment chain on the right is generally not guaranteed if $\mathbf{Y}^{n+1} \notin \mathcal{R}$. Then the dual relation (12) implies that $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ is guaranteed to cover the unknown test outcome $\mathbf{Y}^{n+1}$ whenever $\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1})$ and $\mathbf{Y}^{n+1} \in \mathcal{R}$ hold simultaneously.

Of course, we do not have the extra information on which fixed region $\mathcal{R}$ contains $\mathbf{Y}^{n+1}$ in practice, but if we can find a sufficiently rich collection of regions $\mathcal{P}$ such that $\mathbf{Y}^{n+1}$ belongs to one of these regions with high probability, we can obtain a valid prediction set by aggregating the local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ over all $\mathcal{R} \in \mathcal{P}$. Furthermore, to ensure interpretability, we report a relatively tight rectangular set that bounds the union of all localized prediction sets, rather than the union itself.

Thanks to our previous GWC construction, we already have a valid candidate for such high-probability regions, namely the global-worst-case prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Let $\mathcal{P}$ be a *random partition* of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. In summary, we construct $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ so that it encloses the union of all local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ for $\mathcal{R} \in \mathcal{P}$ with a rectangle, while remaining contained in the global-worst-case prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Letting $\text{Rect}_{\mathcal{P}}(A)$ denote a rectangle that contains the set A while being contained in $\bigsqcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R} = \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$, we define the local-worst-case prediction set as

$$\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) := \text{Rect}_{\mathcal{P}}\left(\bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \right) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}). \tag{20}$$

In practice, our method constructs a *relatively tight*, though not necessarily the *tightest*, such rectangle due to computational considerations. Nonetheless, the resulting prediction sets are (often much) more informative than $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ in finite samples. Moreover, in the large-sample limit, the union in (20) becomes approximately rectangular itself, making this enclosing step a very mild relaxation from the perspective of statistical efficiency.

The following result guarantees that this high-level approach leads to prediction set $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ that retains valid marginal coverage.

Proposition 6 *Consider a (random) collection $\mathcal{P}$ of disjoint regions $\mathcal{R} \subseteq \mathbb{R}^d$ satisfying $\sqcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R} = \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ almost surely. For any $\mathcal{R} \in \mathcal{P}$, let $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \subseteq \mathbb{R}^d$ denote a local prediction set satisfying* (19) *almost surely. Then, $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ defined in* (20) *satisfies $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Moreover, if $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ are exchangeable and Assumption 1 holds, $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})] \geq 1 - \alpha$.*

We next introduce an intuitive data-driven partition based on the possible rankings of the test residuals E_j^{n+1} relative to the calibration residuals $(E_j^1, \dots, E_j^n)$ for each coordinate $j \in [d]$, and then we characterize the corresponding prediction set $\hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$.

3.4.2 Step 1: A Data-Driven Partition Based on Sample Order Statistics

For each $j \in [d]$, let $E_j^{(k)}$ denote the k-th order statistic of $\{E_j^i\}_{i=1}^n$, so that $0 := E_j^{(0)} < E_j^{(1)} \leq \dots \leq E_j^{(n)} < E_j^{(n+1)} := +\infty$. For each multi-index $h = (h_1, \dots, h_d) \in [n+1]^d$, define the local region $\mathcal{Y}^h$ in the outcome space as:

$$\begin{aligned} \mathcal{Y}^h &:= \left\{ \mathbf{y} \in \mathbb{R}^d : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{E}^h \right\}, \quad \text{where} \\ \mathcal{E}^h &:= \bigtimes_{j=1}^d \left[E_j^{(h_j - 1)}, U_j^{h_j} \right) \quad \text{and} \quad U_j^{h_j} := \min\left\{ E_j^{(h_j)}, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha}) \right\}. \end{aligned} \tag{21}$$

Clearly $\mathcal{P}_{n,d} = \{\mathcal{Y}^h\}_{h \in [n+1]^d}$ partitions $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ into $(n+1)^d$ disjoint regions, some of which may be empty depending on $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$. Figure 2 visualizes this partition and the corresponding construction of $\hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$, in a two-dimensional toy example.

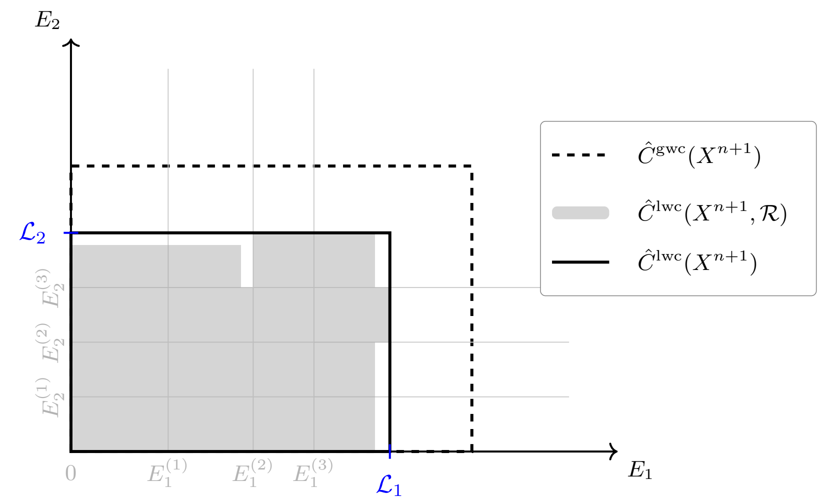

Figure 2: Illustration of the construction of transductively standardized conformal prediction sets using the method outlined in (20), with the data-driven partition $\mathcal{P}_{n,d}$ of the outcome space defined in (21), in a toy example with a two-dimensional outcome variable. The gray grid lines indicate the partition $\mathcal{P}_{n,d}$ in the residual space, as determined by the order statistics $E_j^{(k)}$ of the calibration residuals. The shaded rectangles represent local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ corresponding to different hypothesized rectangles $\mathcal{R} \in \mathcal{P}_{n,d}$. The solid rectangle shows the projection, in residual space, of their rectangular enclosure defined in (20), which constitutes our final prediction set. For comparison, the larger dashed rectangle shows the corresponding projection of the more conservative prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ from Section 3.3.

While it may appear at first sight that this choice of partition necessarily incurs exponential computational costs of order $\mathcal{O}(n^d)$, rendering it intractable for all but very small dimensions d, in fact it enables a much more efficient characterization of the local prediction sets $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1};\mathcal{Y}^h)$ that will lead to a practical algorithm for constructing $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$ at a computational cost of only $\mathcal{O}(d^2 n\log n)$.

3.4.3 Step 2: Characterization of the Local Prediction Sets

For any $h\in[n+1]^d$ with $\mathcal{Y}^h\neq\emptyset$, we characterize the local prediction set $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1};\mathcal{Y}^h)$ by computing the scalar quantity $\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^h)$ that satisfies the constraint (17), following a strategy similar to the worst-case approach from Section 3.3.

Define the LWC scalarization map $\hat{\Phi}^{\mathrm{lwc}}_h:\mathbb{R}^d_+\to\mathbb{R}$ such that

$$\hat{\Phi}^{\mathrm{lwc}}_h(\mathbf{t}):=\max_{1\le j\le d}\left\{\sup_{\mathbf{z}\in\mathcal{E}^h}\frac{t_j}{\hat{\sigma}_j(z_j)}\;-\;\inf_{\mathbf{z}\in\mathcal{E}^{\mathrm{gwc}}}\frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}\right\}\tag{22}$$

where $\mathcal{E}^{\mathrm{gwc}}:=\bigtimes_{j=1}^d\left[0,\,\omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})\right]$ is the projection of $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ onto the residual space that satisfies $\mathbf{y}\in\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})\iff\mathbf{E}(\mathbf{X}^{n+1},\mathbf{y})\in\mathcal{E}^{\mathrm{gwc}}$. With this choice, we have

$$\mathbf{E}^{n+1}\in\mathcal{E}^h\implies\hat{\Phi}^{\mathrm{lwc}}_h(\mathbf{t})\;\ge\;\max_{1\le j\le d}\sup_{\mathbf{z}\in\mathcal{E}^h}\frac{t_j-\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}\ge\Phi^{\mathrm{oracle}}(\mathbf{t}),\quad\text{for all }\mathbf{t}\in\mathbb{R}^d_+.\tag{23}$$

By (21), this ensures that for any $h\in[n+1]^d$ with $\mathbf{Y}^{n+1}\in\mathcal{Y}^h\neq\emptyset$, $\hat{\Phi}^{\mathrm{lwc}}_h\ge\Phi^{\mathrm{oracle}}$ uniformly. Moreover, $\hat{\Phi}^{\mathrm{lwc}}_h(\mathbf{t})$ admits the following closed form.

Lemma 7 *If $\mathcal{Y}^h\neq\emptyset$, then $\hat{\Phi}^{\mathrm{lwc}}_h(\mathbf{t})$ defined in (22) can be written for any $\mathbf{t}\in\mathbb{R}^d_+$ as:*

$$\hat{\Phi}^{\mathrm{lwc}}_h(\mathbf{t})=\max_{1\le j\le d}\left\{\frac{t_j}{r_j(h_j)}-\min\left\{\frac{\hat{\mu}_j(0)}{\hat{\sigma}_j(0)},\;\lim_{z_j\uparrow\omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})}\frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}\right\}\right\},$$

where $r_j(h_j)=\hat{\sigma}^{\mathrm{obs}}_j$ if $E_j^{(h_j-1)}\le\hat{\mu}^{\mathrm{obs}}_j<U_j^{h_j}$, and otherwise $\min\left\{\hat{\sigma}_j(E_j^{(h_j-1)}),\hat{\sigma}_j(U_j^{h_j})\right\}$.

Note that Lemma 7 involves a limit for $z_j\uparrow\omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$ because $\omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$ can be infinite when $\hat{Q}^{\mathrm{gwc}}_{1-\alpha}\ge\frac{n}{\sqrt{n+1}}$ (recall Lemma 2), in which case $\mathcal{E}^{\mathrm{gwc}}=\mathbb{R}^d_+$. Nonetheless, our construction remains well-defined even in that special case.

Using Lemma 7, we compute $S^i_{\mathrm{lwc}}(\mathcal{Y}^h)=\hat{\Phi}^{\mathrm{lwc}}_h(\mathbf{E}^i)$ for all $i\in[n]$ and define $\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^h):=\hat{Q}_{1-\alpha}(S^1_{\mathrm{lwc}}(\mathcal{Y}^h),\dots,S^n_{\mathrm{lwc}}(\mathcal{Y}^h))$. Under the working hypothesis $\mathbf{Y}^{n+1}\in\mathcal{Y}^h$, it follows immediately from (23) that $\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^h)\ge\hat{Q}^{\mathrm{oracle}}_{1-\alpha}$ almost surely. By substituting $c=\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^h)$ in (11), we obtain the local prediction set $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1};\mathcal{Y}^h)$ in (18) with $\mathcal{R}=\mathcal{Y}^h$ and it satisfies (19).

The procedure above summarizes the construction when $\mathcal{Y}^h\neq\emptyset$. If $\mathcal{Y}^h=\emptyset$, we can simply set $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1};\mathcal{Y}^h)=\emptyset$ and the containment (19) holds trivially. Algorithm 1 summarizes this construction.

Algorithm 1 Construction of local prediction sets under working hypothesis $\mathbf{Y}^{n+1} \in \mathcal{Y}^h$

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; global bounds $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$; calibration mean $\hat{\mu}^{\text{obs}}$ and standard deviation $\hat{\sigma}^{\text{obs}}$; index $h \in [n+1]^d$.

1: **if** $\mathcal{Y}^h \neq \emptyset$ **then**
2: Compute $S^i_{\text{lwc}}(\mathcal{Y}^h) \leftarrow \hat{\Phi}^{\text{lwc}}_h(\mathbf{E}^i)$ **for** $i \in [n]$ using $\hat{\Phi}^{\text{lwc}}_h$ defined in Lemma 7.
3: Compute $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h) \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{lwc}}(\mathcal{Y}^h), \dots, S^n_{\text{lwc}}(\mathcal{Y}^h))$.
4: **return** $\omega_1(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h)), \dots, \omega_d(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h))$ using ω defined in Lemma 2.
5: **else**
6: **return** $\emptyset$.
7: **end if**

3.4.4 Computational Shortcut for Rectangular Enclosure

While the characterization provided above makes it easy to compute $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)$ for any specific index $h \in [n+1]^d$, the problem remains that the intended construction of $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ in (20) involves an *exponential* number of such local prediction sets.

Therefore, to obtain a practical method, we must now develop a shortcut that avoids full enumeration of all $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)$. Our solution begins by noting that $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)$ from (18) can be re-written as:

$$\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h) = \bigcap_{j=1}^d \left\{ \mathbf{y} \in \mathbb{R}^d : E_j^{(h_j-1)} \leq E_j(\mathbf{X}^{n+1}, y_j) \leq B_j^h \right\} \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}),$$

where the coordinate-wise upper bounds B_j^h are given by

$$B_j^h := \begin{cases} \min\left\{ U_j^{h_j},\ \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h)) \right\}, & \text{if } U_j^{h_j} > E_j^{(h_j-1)} \text{ and } \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h)) > E_j^{(h_j-1)}, \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

For every coordinate $j \in [d]$, define the j-th residual boundary $\mathcal{L}_j := \max_{h \in [n+1]^d} B_j^h$. Then the rectangular enclosure of all local prediction sets in (20) can be expressed as

$$\begin{aligned} \text{Rect}_{\mathcal{P}_{n,d}}\left(\bigsqcup_{h \in [n+1]^d} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h) \right) &= \bigcap_{j=1}^d \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \mathcal{L}_j \right\} \\ &=: \hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1}). \end{aligned} \tag{25}$$

Notably, this rectangular enclosure is typically very tight in practice, because the union of the local prediction sets constructed using our designated partition $\mathcal{P}_{n,d}$ is itself very close to being a rectangle when the sample size is large enough.

The last remaining step to make our method practical is finding an efficient algorithm for computing the boundaries $\mathcal{L}_j$ for all $j \in [d]$, without requiring enumeration of all local prediction sets. Such a shortcut is made possible by the following result, which simplifies the definition of $\mathcal{L}_j$ from a maximum over $(n+1)^d$ distinct values to a maximum over an explicit, narrowed-down list of only $n+1$ relevant candidates.

Before stating this result, it is helpful to introduce some notation. Define the (random) *mean index* vector $h^* \in [n+1]^d$ as the unique index such that $E_j^{(h_j^*-1)} \leq \hat{\mu}_j^{\text{obs}} \leq E_j^{(h_j^*)}$ for all $j \in [d]$. Let $\text{Row}_j(h^*)$ be a subset of $[n+1]^d$ containing exactly $n+1$ indices, defined as those coinciding with the mean index h^* on all coordinates other than the j-th one:

$$\text{Row}_j(h^*) := \left\{ h \in [n{+}1]^d : h_k = h_k^* \text{ for all } k \neq j \right\}, \quad \forall j \in [d].$$

Lemma 8 *If $\mathcal{Y}^{h^*} \neq \emptyset$, then for any $j \in [d]$, then $\mathcal{L}_j := \max_{h\in[n+1]^d} B_j^h$ can be equivalently written as:*

$$\mathcal{L}_j = \max_{h\in\text{Row}_j(h^*)} B_j^h = B_j^{h^{[j]}} > 0, \quad \textit{where } h^{[j]} := \underset{h\in\text{Row}_j(h^*)}{\text{argmax}} B_j^h.$$

This maximum index $h^{[j]}$ satisfies $B_j^h = 0$ for any $h \in \text{Row}_j(h^*)$ with $h_j \geq h_j^{[j]}$.

Moreover, for every $j \in [d]$, if there exists $m \in \text{Row}_j(h^*)$ with $m_j \geq h_j^*$ and $B_j^m = 0$, then $B_j^h = 0$ for all $h \in \text{Row}_j(h^*)$ with $h_j \geq m_j$.

Under the regularity condition $\mathcal{Y}^{h^*} \neq \emptyset$, which can be immediately verified and typically holds in practice under all but the most extreme scenarios, Lemma 8 gives a very fast algorithm for computing $\mathcal{L}_j$ separately for each coordinate $j \in [d]$. In particular, one can find $h^{[j]}$, where only $h_j^{[j]}$ needs to be determined, by performing a simple backward search over $h \in \text{Row}_j(h^*)$ with $h_j = n+1, \ldots, 1$ and stopping for the first h_j with $B_j^h > 0$. Moreover, if $h_j^{[j]} \geq h_j^*$, which can be determined by simply checking whether $B_j^{h^*} > 0$, we can find $h^{[j]}$ even more effectively through a binary search that checks whether $B_j^m = 0$ at each intermediate split step $m \in \text{Row}_j(h^*)$ with $m_j \in [h_j^*, ..., n+1]$ and assigns $h_j^{[j]}$ to be the largest m_j such that $B_j^m > 0$.

The special case $\mathcal{Y}^{h^*} = \emptyset$ corresponds to an extreme situation where even the conservative GWC prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ does not contain the empirical mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ of the calibration residuals. Typically, this should not occur at moderate miscoverage levels α, unless the data are extremely heavy-tailed. Such cases can be handled by simply falling back to the conservative GWC prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ itself, which we know to have valid marginal coverage. In conclusion, the procedure described above, summarized in Algorithm 2, produces:

$$\hat{C}(\mathbf{X}^{n+1}) = \begin{cases} \hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1}), & \text{if } \mathcal{Y}^{h^*} \neq \emptyset, \\ \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}), & \text{otherwise.} \end{cases} \tag{26}$$

Theorem 9 *Assume $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ are exchangeable and Assumption 1 holds. Then, $\hat{C}(\mathbf{X}^{n+1})$ constructed by Algorithm 2 satisfies: $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \geq 1-\alpha$.*

Proof Using the characterization of $\hat{C}(\mathbf{X}^{n+1})$ in (26), the proof follows immediately by combining Proposition 6, which tells us that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})] \geq 1-\alpha$, and Theorem 5, which tells us that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})] \geq 1-\alpha$. ■

Theorem 9 guarantees that our prediction sets $\hat{C}(\mathbf{X}^{n+1})$ defined in (26) have valid finite-sample marginal coverage above the target level $1-\alpha$. Moreover, these prediction sets have an interpretable rectangular shape and are typically much tighter than $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$.

Algorithm 2 Transductively Standardized Conformal Prediction (TSCP)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

Compute $\hat{\mu}, \hat{\sigma}$ using (14).
Compute global bounds $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$ using TSCP-GWC. ▷ Algorithm S5.
Sort $\{E_j^i\}_{i=1}^n$ for each $j \in [d]$ to obtain order statistics $E_j^{(k)}$.
Compute $\mathcal{Y}^{h^*}$ using (21).
if $\mathcal{Y}^{h^*} = \emptyset$ **then**
    return $\hat{C}(\mathbf{X}^{n+1}) \leftarrow \Big\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha}), \quad \forall j \in [d]\Big\}$.
else
    for $j = 1$ to d **do**
        Compute $\omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^{h^*}))$ using Algorithm 1 and $B_j^{h^*}$ using (24).
        if $B_j^{h^*} = 0$ **then**
            Backward search over $h_j = h_j^* - 1, \dots, 1$ to find $h^{[j]}$. ▷ Algorithm S6
        else
            Binary search over $h_j = h_j^*, \dots, n+1$ to find $h^{[j]}$. ▷ Algorithm S7
        end if
        Update $\mathcal{L}_j \leftarrow B_j^{h^{[j]}}$.
    end for
    return $\hat{C}(\mathbf{X}^{n+1}) \leftarrow \Big\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \mathcal{L}_j, \quad \forall j \in [d]\Big\}$.
end if

The computational complexity of Algorithm 2 ranges from $\mathcal{O}(d^2 n \log n)$ in the best case, when binary searches can be used for all coordinates, to $\mathcal{O}(d^2 n^2)$ in the worst case, when all coordinates require backward searches; see Appendix A.3 for details. Even in the worst case, a cost of $\mathcal{O}(d^2 n^2)$ is typically not prohibitive in practice: calibration sample sizes n for conformal prediction are usually in the hundreds to low thousands, and beyond that additional samples are often better spent improving model training.

In addition, the computation of the distinct boundaries $\mathcal{L}_1, \dots, \mathcal{L}_d$ is trivially parallelizable, and the entire procedure need not be repeated across different test inputs, since $\mathcal{L}_1, \dots, \mathcal{L}_d$ are independent of $\mathbf{X}^{n+1}$—remarkably, we have arrived through *transductive* logic at a method that is effectively *inductive*. Therefore, applying Algorithm 2 to m different test points using the same calibration data set has total cost ranging from $\mathcal{O}(d^2 n \log n + dm)$ to $\mathcal{O}(d^2 n^2 + dm)$ for typical choices of generalized residuals (Section 2.3).

4 Numerical Experiments

We evaluate Transductively Standardized Conformal Prediction (TSCP) on simulated and real multivariate regression problems. The experiments are organized around four questions. First, does TSCP retain joint marginal coverage while adapting the interval length of each outcome coordinate? Second, is the conclusion stable under dependence and partial heteroskedasticity? Third, does the advantage persist with quantile-regression scores and against CQHR? Finally, what is the observed wall-clock cost of the construction? Additional sensitivity analyses, including heavy tails, contamination, and a convex shape-template baseline, are reported in Appendix C. The original experiments are retained in Appendix D, including the original Gaussian comparisons, distribution-specific tables, and comparisons with the oracle and unenclosed local union. Those historical results retain their original resampling protocol and are distinguished from the fresh-draw experiments below.

4.1 Methods and Evaluation Metrics

The primary comparisons use four methods. *Unscaled Max* applies split conformal prediction to the maximum coordinate-wise residual. *Point CHR* estimates coordinate scales using an additional data split. *Emp. Copula* estimates a dependence model from the calibration residuals and then uses the same observations for calibration; this implementation does not have the finite-sample guarantee enjoyed by the other methods. *TSCP* denotes the rectangular local-worst-case construction proposed in this paper. For quantile-regression experiments, we additionally compare with the native CQHR method of Sampson and Chan (2024). Appendix C.8 contains a separate low-dimensional comparison with the hyperrectangular shape-template method of Tumu et al. (2024).

For a test sample of size m, we report the empirical joint coverage

$$\widehat{\mathrm{Cov}} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{1}\Big\{\boldsymbol{Y}_i \in \widehat{C}(\boldsymbol{X}_i)\Big\}.$$

For absolute residual scores, all methods return a rectangle with residual half-widths $(\mathcal{L}_1, \dots, \mathcal{L}_d)$. We therefore report the normalized residual-space volume $\prod_{j=1}^{d} \mathcal{L}_j$, which equals 2^{-d} times the full rectangular volume, and the full coordinate-wise interval lengths $2\mathcal{L}_j$. For CQR, the base interval depends on the test covariates, so volume and coordinate lengths are averaged in the original outcome space. Runtime is the wall-clock time needed to construct the conformal region after residuals have been computed; fitting the common regression model is excluded. Unless stated otherwise, absolute-residual results are averaged over 200 repetitions and the target coverage is $1 - \alpha = 0.9$. Every synthetic repetition redraws independent training, test, and calibration observations while holding the underlying regression coefficients fixed. The figures show mean curves without error bars; across-repetition standard deviations and an uncertainty audit are reported in Appendix C.1 and the accompanying CSV files.

4.2 Absolute-Residual Experiments

We first generate 10-dimensional features $\boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, I_{10})$ and outcomes

$$Y_j = f_j(\boldsymbol{X}) + \epsilon_j, \qquad j \in [10],$$

where every f_j is linear and the nonzero coefficients are sampled once from $\mathrm{Unif}(-10, 10)$. Each repetition independently draws 7,200 observations to fit ordinary least squares, 800 test observations, and a separate calibration sample of size $n \in \{30, 50, 100, 300, 500\}$.

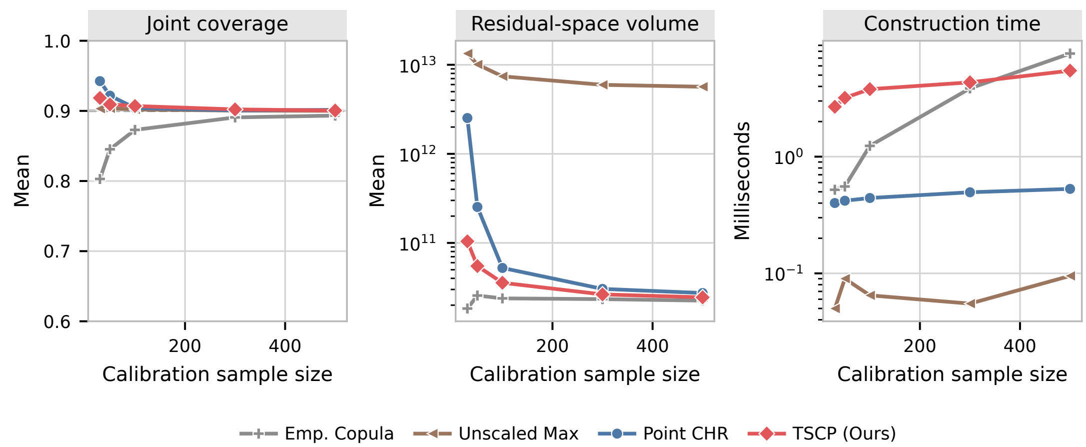

Figure 3: Absolute-residual experiment with $d = 10$, heterogeneous Gaussian scales $(10, 9, \dots, 1)$, and equicorrelation $\rho = 0.6$. Curves report averages over 200 repetitions; the dashed line marks the target joint coverage. Runtime excludes the common model fit.

Heterogeneous and dependent noise. The first experiment uses $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, DRD)$, where $D = \mathrm{diag}(10, 9, \dots, 1)$ and R is equicorrelated with $\rho = 0.6$. This simultaneously tests heterogeneous coordinate scales and cross-output dependence. Figure 3 reports joint coverage, residual-space volume, and construction time; the coordinate-wise lengths are reported separately in Figure 4. At $n = 100$, TSCP attains coverage 0.907 with average volume 3.57×10^{10}, compared with coverage 0.904 and volume 5.24×10^{10} for Point CHR. Unscaled Max has coverage 0.902 but a volume of 7.42×10^{12}. Emp. Copula is smaller but undercovers, with coverage 0.873.

Figure 4 makes the dimension-specific behavior explicit at $n = 100$. The mean TSCP full interval length decreases from 49.4 on the noisiest coordinate to 5.0 on the quietest coordinate. In contrast, Unscaled Max assigns the same length, approximately 38.0, to every coordinate. Thus, similar overall coverage does not imply similar marginal informativeness: TSCP allocates width according to coordinate scale instead of allowing the noisiest output to determine all ten intervals. The smaller Emp. Copula bars must be read together with its joint undercoverage in Figure 3.

Partial heteroskedasticity. To distinguish heterogeneity across coordinates from heteroskedasticity with respect to the input, we next make only the first five coordinates heteroskedastic. For $j \leq 5$, conditional on $X_1 = x_1$, the noise standard deviation is

$$\sigma_j(x_1) = (11 - j)\sqrt{\frac{1 + 1.5x_1^2}{2.5}},$$

while $\sigma_j(x_1) = 11 - j$ for $j > 5$. The normalization preserves $\mathbb{E}[\sigma_j^2(X_1)] = (11 - j)^2$, so the comparison isolates input-dependent variance rather than changing the marginal second moments.

Figure 5 shows that TSCP remains close to the target across calibration sizes and continues to adapt coordinate-wise. At $n = 100$, TSCP obtains coverage 0.904 and average volume 1.40×10^{11}; Point CHR obtains coverage 0.902 and volume 2.94×10^{11}, while Unscaled

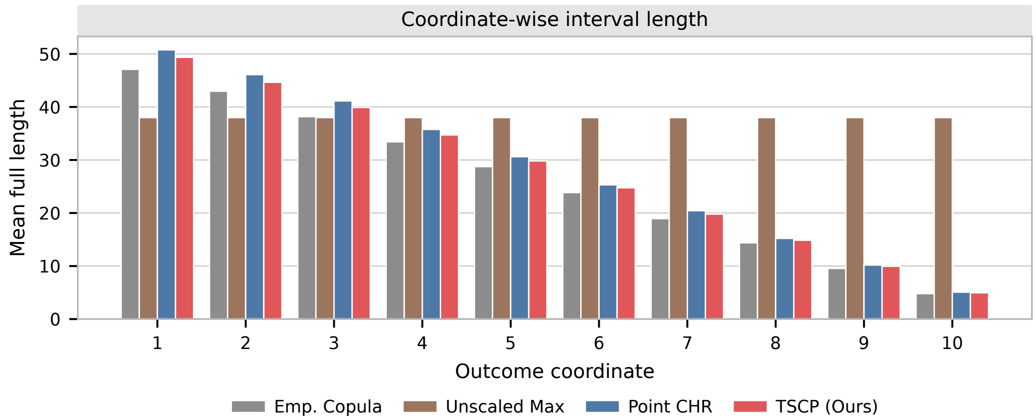

Figure 4: Coordinate-wise mean full interval lengths for the absolute-residual experiment with $d = 10$, $n = 100$, heterogeneous Gaussian scales $(10, 9, \ldots, 1)$, and equicorrelation $\rho = 0.6$. Residual-space half-widths are multiplied by two to report full outcome intervals. Results average 200 independent repetitions.

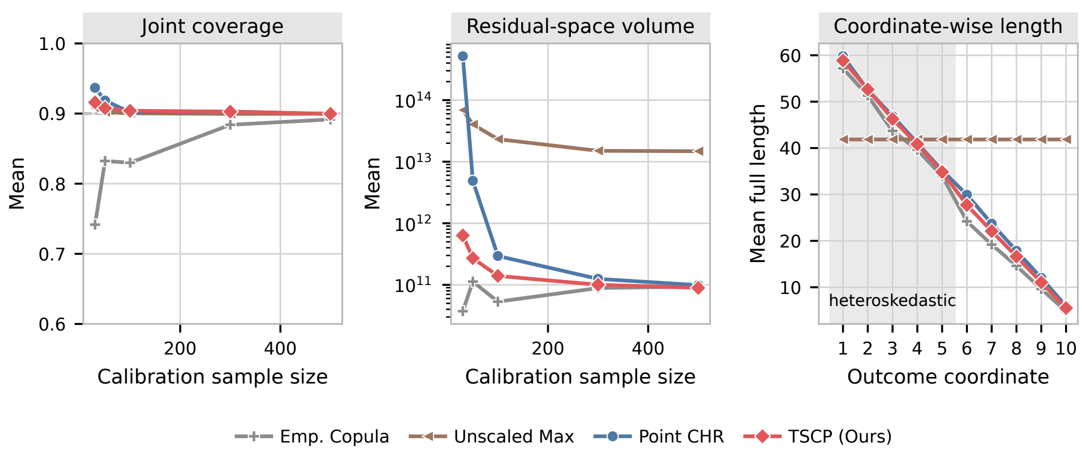

Figure 5: Performance when only the first five of ten outcomes are heteroskedastic in X_1. The shaded coordinates in the right panel are the heteroskedastic ones; full interval lengths are shown for $n = 100$. Results are averaged over 200 independent repetitions.

Max obtains coverage 0.900 and volume 2.33×10^{13}. This is a marginal-coverage experiment. It does not assert conditional coverage at each value of X_1, nor does the present global coordinate standardization attempt to estimate the local variance function.

4.3 Quantile-Regression Scores and CQHR

We next replace absolute mean-regression residuals with coordinate-wise CQR scores. Let $\widehat{q}_{\ell,j}(x)$ and $\widehat{q}_{u,j}(x)$ denote fitted lower and upper conditional quantiles. The signed score is

$$S_j(x, y) = \max\{\widehat{q}_{\ell,j}(x) - y_j, y_j - \widehat{q}_{u,j}(x)\}.$$

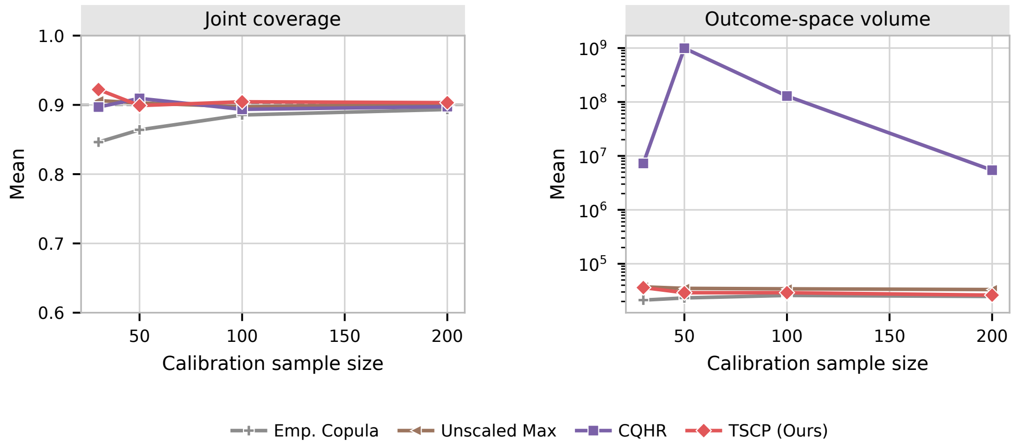

Figure 6: CQR experiment with $d = 3$ and base-interval miscoverage 0.5. TSCP uses capped nonnegative scores, Unscaled Max and Emp. Copula use raw signed scores, and CQHR uses its native quantile-hyperrectangle construction. Results are averaged over 30 independent repetitions.

Negative values correspond to observations strictly inside the base interval. Because the current TSCP theory is stated for nonnegative scores, the primary experiment uses the capped score $E_j = \max\{S_j, 0\}$. For a nonnegative coordinate threshold, this transformation yields an expanded base interval and avoids turning a negative signed score into a positive error magnitude, as an absolute-value transformation would. It cannot shrink an already conservative base interval. Capping also creates ties at zero, so this experiment does not satisfy the pairwise-distinctness condition of Assumption 1 verbatim. We report its empirical performance without claiming an extension of the theorem to all tied or degenerate residual configurations. Unscaled Max and Emp. Copula use the raw signed scores, while CQHR is applied in its native form. Predicted lower and upper quantiles are ordered pointwise for all methods. CQHR uses the first coordinate as reference and a numerical floor of 10^{-12} on base-interval lengths when forming width ratios.

We simulate $d = 3$ Gaussian outcomes with noise scales $(3, 2, 1)$, fit separate gradient-boosted quantile regressions using 2,400 training observations, and evaluate on 600 test observations. Each quantile model uses 100 boosting stages, maximum tree depth 5, learning rate 0.05, and minimum leaf size 5. The base-interval miscoverage is 0.5 and the final target miscoverage is 0.1. Each of the 30 repetitions uses fresh training, test, and calibration samples. Figure 6 shows that TSCP's empirical coverage fluctuates around the target, ranging from 0.899 to 0.922 over the displayed calibration sizes, while its mean outcome-space volume remains between 2.62×10^4 and 3.62×10^4. CQHR attains empirical coverage between 0.894 and 0.909 but is much larger and more variable in this design; at $n = 100$, its mean volume is 1.29×10^8. This comparison is specific to the stated base quantile models and is not a claim that one method uniformly dominates CQHR. Base-interval and shifted-score sensitivity analyses are in Appendix C.7.

Figure 7 reports the corresponding coordinate-wise comparison at $n = 100$. TSCP's mean full lengths are $(36.2, 36.9, 20.9)$, compared with $(33.3, 33.8, 29.0)$ for Unscaled Max

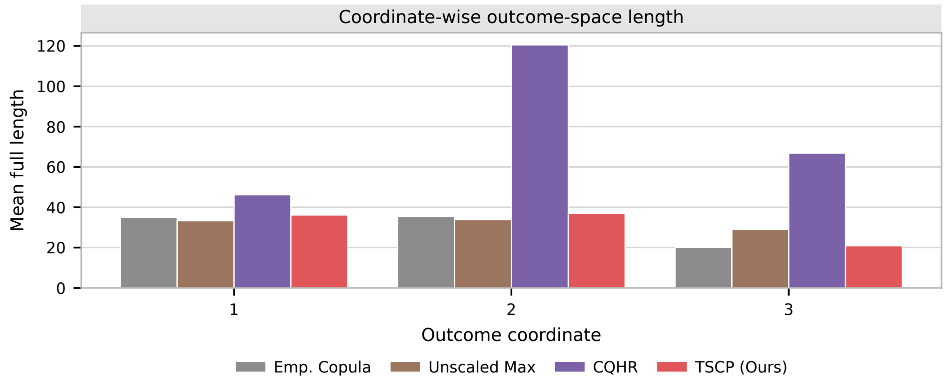

Figure 7: Coordinate-wise mean full outcome-space lengths for the CQR experiment with $d = 3$, $n = 100$, base-interval miscoverage 0.5, and final target coverage 0.9. TSCP uses capped scores, Unscaled Max and Emp. Copula use raw signed scores, and CQHR uses its native construction. Results average 30 independent repetitions.

and $(46.2, 120.5, 66.9)$ for CQHR. In particular, TSCP preserves a substantially shorter third-coordinate interval than Unscaled Max, while CQHR is much wider on the second and third coordinates in this fitted-quantile design. As above, the coordinate-wise lengths should be interpreted together with the joint coverage panel rather than compared without regard to calibration.

4.4 Real Data and Wall-Clock Runtime

We retain the six real-data benchmarks from the original experiments: `rf2` (Tsoumakas et al., 2022), `scm1d` (Contributors, 2019a), `scm20d` (Contributors, 2019b), `stock` (Yeh, 2015), `student` (Cortez, 2008), and `energy` (Tsanas and Xifara, 2012). The table and runtime figure use the order `stock`, `rf2`, `scm1d`, `scm20d`, `energy`, and `student`. Each result averages 200 random training/calibration/test splits using proportions 75%/5%/20%. The calibration sizes therefore range from 15 on `stock` to 490 on `scm1d`. A multitask lasso is used for `stock`; random forests are used for the other datasets. Table 1 reproduces coverage and residual-space volume for the four primary methods.

Figure 8 provides the wall-clock comparison requested by the reviewers. TSCP constructs a region in 0.74–7.46 milliseconds on these datasets. It is slower than the algebraically simpler Unscaled Max and Point CHR implementations, but substantially faster than Emp. Copula on the two 16-dimensional SCM datasets, where Emp. Copula requires approximately 95–118 milliseconds. These measurements include the complete conformal region construction but exclude fitting the shared regression model.

The infinite Point CHR volume on `stock` should not be interpreted as an intrinsic failure of the method. With only $n = 15$ calibration observations, its additional split leaves too few observations for a finite 0.9 conformal quantile. A practitioner committed to Point CHR would allocate a larger fraction of the data to its final calibration stage, at the cost of fewer observations for model and scale estimation.

Dataset	Method	d	Coverage	Volume
`stock`	Unscaled Max	6	0.940 (0.057)	6.03e−2 (1.57e−1)
	Emp. Copula	6	0.727 (0.113)	1.01e−5 (9.80e−6)
	Point CHR	6	1.000 (0.000)	∞
	TSCP	6	0.956 (0.059)	**1.86e−3 (8.55e−3)**
`rf2`	Unscaled Max	8	0.900 (0.016)	4.14e3 (2.90e3)
	Emp. Copula	8	0.893 (0.016)	5.35e1 (4.69e1)
	Point CHR	8	0.901 (0.022)	**6.85e1 (7.55e1)**
	TSCP	8	0.901 (0.017)	5.42e2 (1.50e3)
`scm1d`	Unscaled Max	16	0.900 (0.016)	2.74e39 (4.02e39)
	Emp. Copula	16	0.893 (0.016)	1.09e39 (1.34e39)
	Point CHR	16	0.905 (0.020)	2.84e39 (5.43e39)
	TSCP	16	0.901 (0.015)	**1.52e39 (2.61e39)**
`scm20d`	Unscaled Max	16	0.901 (0.016)	2.70e40 (2.92e40)
	Emp. Copula	16	0.892 (0.017)	1.18e40 (1.19e40)
	Point CHR	16	0.902 (0.023)	3.00e40 (8.00e40)
	TSCP	16	0.901 (0.016)	**1.53e40 (1.50e40)**
`energy`	Unscaled Max	2	0.917 (0.049)	1.58e1 (6.80e0)
	Emp. Copula	2	0.886 (0.061)	5.41e0 (4.49e0)
	Point CHR	2	0.899 (0.069)	8.81e0 (2.42e1)
	TSCP	2	0.922 (0.048)	**6.95e0 (4.43e0)**
`student`	Unscaled Max	3	0.904 (0.055)	**1.51e2 (1.27e2)**
	Emp. Copula	3	0.861 (0.073)	1.08e2 (7.19e1)
	Point CHR	3	0.939 (0.057)	6.89e2 (1.07e3)
	TSCP	3	0.912 (0.056)	1.98e2 (1.77e2)

Table 1: Performance on real datasets, averaged over 200 random splits. One standard deviation is reported in parentheses. Bold volume denotes the smallest average volume among the methods with empirical coverage at or above the target, up to Monte Carlo variation. The calibration sizes are 15, 384, 490, 448, 38, and 32 in the order shown.

Retained coordinate-wise real-data illustration. To emphasize their interpretability, Table 2 displays the rectangular prediction sets produced by different methods for two representative test points from the *energy* dataset, where $d = 2$. In this setting, rectangular prediction sets consist of pairs of simultaneous prediction intervals that aim to jointly contain the true outcomes with probability 90%.

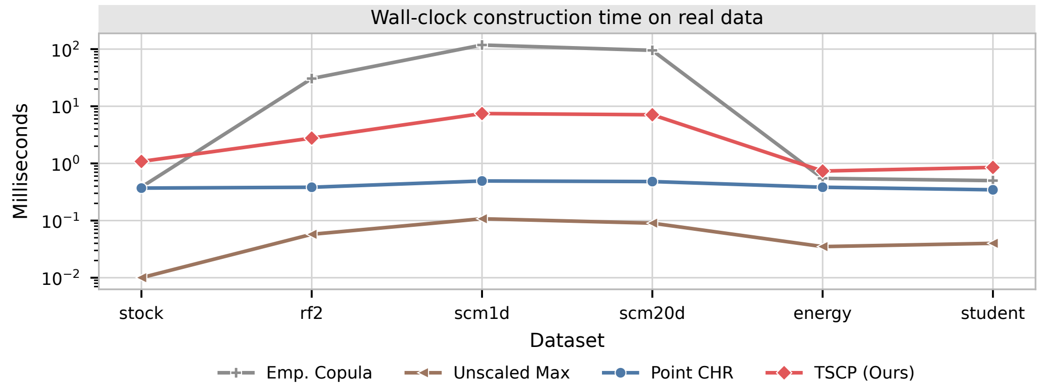

Figure 8: Average wall-clock time for conformal region construction on the six real datasets. The vertical axis is logarithmic and reports milliseconds. All methods use the same fitted regression model in each split; model fitting and residual computation are excluded.

Test 1/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[31.70, 38.94]	**[34.74, 41.98]**	13.09	(35.40, 39.22)
Emp. copula	**[34.36, 36.28]**	**[34.74, 41.98]**	3.47	
Point CHR	[34.29, 36.36]	[31.45, 45.27]	7.14	
TSCP	[34.28, 36.36]	[33.63, 43.08]	4.93	

Test 2/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[25.44, 32.67]	[29.34, 36.58]	13.09	(29.88, 28.31)
Emp. copula	**[28.10, 30.02]**	[29.34, 36.58]	3.47	
Point CHR	[28.02, 30.09]	[26.05, 39.87]	7.14	
TSCP	[28.01, 30.10]	**[28.24, 37.69]**	4.93	

Table 2: Anecdotal illustration of conformal prediction sets of rectangular shape for two test points from the *energy* dataset. In this case, the prediction sets can be reported as pairs of joint prediction intervals for the two outcome dimensions. Volumes are computed in residual space. Intervals that fail to cover the corresponding true outcomes are highlighted in red; smallest interval that covers the ground truth is highlighted in **bold**. This example is included for illustration only: the coverage guarantees of conformal prediction hold on average, not for individual test points.

4.5 Follow-Up Real-Data Diagnostics

We conduct an additional diagnostic study on all six datasets to measure coordinate-wise informativeness and the frequency of the computational branches in TSCP. We rerun 200 splits per dataset with the original 75%/5%/20% allocation, regression hyperparameters, and deterministic split-seed convention. All four methods share the fitted model and the same calibration and test observations within each split. These are new reruns under the recorded software environment, not recovered original fitted models; they complement the historical measurements in Table 1. Exact split indices and residual matrices are cached for reproducibility.

Coordinate-wise interval lengths. Let $\ell_{j,m}^{(r)} = 2\mathcal{L}_{j,m}^{(r)}$ be the full outcome-space interval length for coordinate j, method m, and split r. To compare outputs with different units, Figure 9 reports the paired length ratio

$$A_{j,m} = \frac{1}{200} \sum_{r=1}^{200} \frac{\ell_{j,m}^{(r)}}{\ell_{j,\text{Unscaled}}^{(r)}}.$$

Thus, a bar below one indicates a shorter interval on average relative to Unscaled Max on the same splits. Ratios are formed before averaging; they are not ratios of mean interval lengths. Raw full lengths, their across-split standard deviations, and coordinate labels are retained in the supporting data. The infinite Point CHR intervals on `stock` are explicitly marked, not truncated or omitted from the numerical summaries.

The profiles show width reallocation rather than uniform shortening. On `energy`, TSCP's mean full lengths are $(2.72, 9.85)$, compared with $(7.77, 7.77)$ for Unscaled Max; the paired ratios are $(0.36, 1.28)$. The corresponding joint coverages are 0.923 and 0.917. On `stock`, TSCP's ratios range from 0.20 to 1.36, and on the two SCM datasets they range from about 0.74 to 1.24. By contrast, on `student` all three TSCP ratios exceed one, namely $(1.01, 1.08, 1.18)$, confirming the retained example where Unscaled Max is more efficient. The heterogeneous `rf2` profile likewise does not erase Point CHR's volume advantage. Marginal and joint coverage for the same reruns are reported in Appendix C.9.

Search regimes and construction cost. We instrument the executed branches of the shortcut algorithm. A fallback occurs when the mean cell has empty intersection with the global rectangle; a backward-search run is a non-fallback run that enters the backward branch for at least one coordinate. We also record the number of coordinates using that branch and the number of candidate cells inspected. Split-level rates use all 200 splits, whereas coordinate-level rates exclude fallback runs.

Table 3 reports these counts and conditional construction times. Timings exclude fitting, residual computation, and diagnostic logging. After a warm-up, we take the median of seven uninstrumented calls on each calibration set and average those medians within each observed regime. Calls are timed serially, after all parallel model fits have finished, with native thread pools limited to one thread. An unobserved regime has no estimable conditional time and is marked by a dash, rather than a zero.

At the main target $\alpha = 0.1$, neither backward search nor fallback occurs in any of the 200 splits for any dataset. All 10,200 coordinate searches use the binary branch, with about 4.00–8.49 candidate-cell inspections per coordinate on average across datasets. The mean construction times range from 0.466 to 7.239 ms. Observing zero events does not establish that their probability is zero: for either split-level event on a given fixed dataset, zero occurrences in 200 randomized splits give a pointwise one-sided 95% binomial upper limit of $1 - 0.05^{1/200} \simeq 1.49\%$. This limit is not simultaneous across datasets and does not treat coordinates within a split as independent. Appendix C.10 examines more extreme targets where both exceptional branches are observed. These measurements describe the tested implementation and do not improve its worst-case complexity bound.

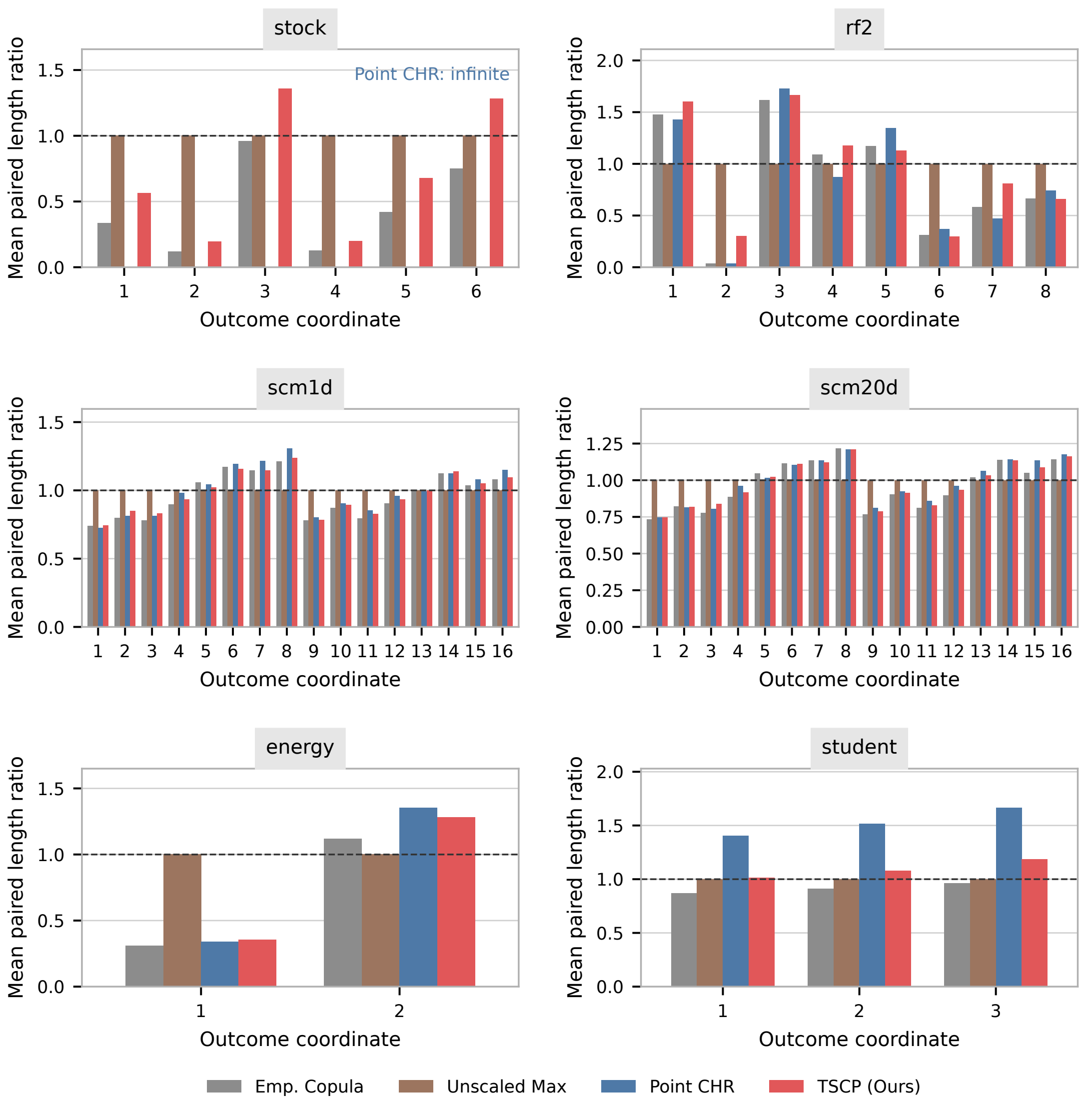

Figure 9: Coordinate-wise full interval lengths on all six real datasets, normalized within each split by the corresponding Unscaled Max length and then averaged over 200 splits. The dashed line marks a ratio of one. Point CHR is infinite on `stock`. These width comparisons should be read with the matching joint coverage in Table S4 and the marginal coverage in Figure S10, not as an unconditional ranking of calibrated performance.

Dataset	n	Backward splits	Backward coordinates	Fallback splits	Binary only (ms)	Any backward (ms)	Fallback (ms)
stock	15	0/200	0/1200	0/200	1.007	–	–
rf2	384	0/200	0/1600	0/200	2.780	–	–
scm1d	490	0/200	0/3200	0/200	7.239	–	–
scm20d	448	0/200	0/3200	0/200	7.120	–	–
energy	38	0/200	0/400	0/200	0.466	–	–
student	32	0/200	0/600	0/200	0.620	–	–

Table 3: Executed TSCP search regimes at target coverage 0.9 in the new real-data reruns. Each split-count denominator is 200. The coordinate denominator is d times the number of non-fallback splits. Conditional times summarize the full TSCP call, not just the backward loop; a dash denotes an unobserved regime. n is the calibration size.

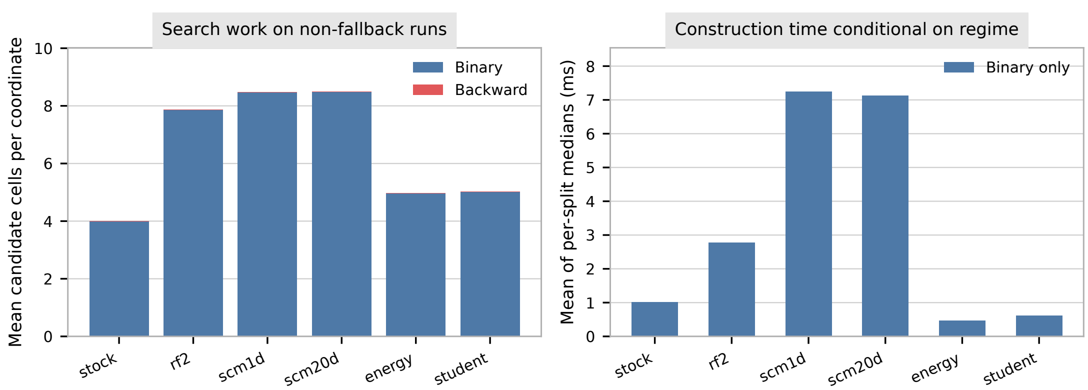

Figure 10: TSCP search work and construction time for the 200 new splits per real dataset at target coverage 0.9. Left: candidate cells inspected per coordinate, averaged over non-fallback runs. Counts include cells rejected by intersection checks. Right: whole-call construction times conditional on the observed regime, using the timing convention in Table 3. Unobserved regimes have no bars.

5 Discussion and Future Work

The core idea underlying the method presented in this paper is the explicit standardization of residuals, which contributes both to the transparency and to the computational efficiency of the resulting prediction sets. As with other standardization-based methods, however, this strategy introduces some sensitivity to outliers. This behavior is visible in the numerical experiments based on the *rf2* dataset, where a small number of influential observations can substantially expand the resulting prediction sets. An important direction for future work is therefore the development of robust extensions of our method, for example through rescaling based on median absolute deviation or quantile-based normalizations, while preserving the exchangeability required for conformal validity.

Additional directions for future work include allowing for negative residuals. Such an extension could, for example, enable shrinkage of baseline prediction sets when starting from quantile regression models that already exhibit conservative coverage. It would also be of interest to adapt the proposed approach to settings with non-numerical outcomes, such as categorical or ordinal responses. Finally, our method could be extended to handle partly observed outcomes (Braun et al., 2025b).

Software Availability

A software implementation of the methods described in this paper, along with code necessary to reproduce the numerical experiments, is available online at `https://github.com/OTheMagic/multi-target-scaling.git`.

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Appendix: Algorithms, Proofs, and Numerical Results

Appendix A. Additional Algorithmic Details

A.1 Implementation of Alternative Methods

Algorithm S1 Separate Prediction Intervals with Bonferroni Correction

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: Compute $\alpha^* \leftarrow \alpha/d$.
2: **for** $j = 1$ to d **do**
3: Compute $\hat{Q}_{1-\alpha^*,j} \leftarrow \hat{Q}_{1-\alpha^*}(E_j^1, \ldots, E_j^n)$.
4: **end for**
5: **return** $\hat{C}^{\text{Bonf.}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha^*,j}, \forall j \in [d]\right\}$.

Algorithm S2 Prediction Intervals via Dimension Reduction with ℓ_∞-Norm (Unscaled)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: Compute $S^i_{\text{unscaled}} \leftarrow \|\mathbf{E}^i\|_\infty$ for $i = 1, \ldots, n$.
2: Compute $\hat{Q}^{\text{unscaled}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{unscaled}}, \ldots, S^n_{\text{unscaled}})$
3: **return** $\hat{C}^{\text{unscaled}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{unscaled}}_{1-\alpha}, \forall j \in [d]\right\}$.

Algorithm S3 Heuristic Plug-In Standardization

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: **for** $j = 1$ to d **do**
2: Compute the location and scale parameters

$$\hat{\mu}_j^{\text{pi}} \leftarrow \frac{1}{n}\sum_{i=1}^n E_j^i, \qquad \hat{\sigma}_j^{\text{pi}} \leftarrow \sqrt{\frac{1}{n-1}\sum_{i=1}^n (E_j^i - \hat{\mu}_j^{\text{pi}})^2}.$$

3: **end for**
4: Compute $S^i_{\text{pi}} \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$ for $i \in [n]$, using Φ defined in (6).
5: Compute $\hat{Q}^{\text{pi}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{pi}}, \ldots, S^n_{\text{pi}})$.
6: Compute

$$\hat{C}^{\text{pi}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{pi}}_{1-\alpha} \cdot \hat{\sigma}_j^{\text{pi}} + \hat{\mu}_j^{\text{pi}}, \forall j \in [d]\right\},$$

which is hyper-rectangular for the standard choices of $\mathbf{E}$ discussed in Section 2.4.
7: **return** Prediction set $\hat{C}^{\text{pi}}(\mathbf{X}^{n+1})$.

Algorithm S4 Standardization with Data Splitting

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.
Output: $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1})$

1: Randomly split $\{1, \dots, n\}$ into two disjoint subsets, $\mathcal{I}_1$ and $\mathcal{I}_2$.
2: **for** $j = 1$ to d **do**
3: Compute the location and scale parameters

$$\hat{\mu}_j^{\text{ds}} \leftarrow \frac{1}{|\mathcal{I}_1|} \sum_{i \in \mathcal{I}_1} E_j^i, \qquad \hat{\sigma}_j^{\text{ds}} \leftarrow \sqrt{\frac{1}{|\mathcal{I}_1| - 1} \sum_{i \in \mathcal{I}_1} (E_j^i - \hat{\mu}_j^{\text{ds}})^2}.$$

4: **end for**
5: Compute $S_{\text{ds}}^i \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}})$ for $i \in \mathcal{I}_2$, using Φ defined in (6).
6: Compute $\hat{Q}_{1-\alpha}^{\text{ds}} \leftarrow \hat{Q}_{1-\alpha}(S_{\text{ds}}^1, \dots, S_{\text{ds}}^n)$.
7: **return** $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha}^{\text{ds}} \cdot \hat{\sigma}_j^{\text{ds}} + \hat{\mu}_j^{\text{ds}}, \forall j \in [d]\right\}$.

A.2 Auxiliary Algorithms

Algorithm S5 Global-Worst-Case TSCP (TSCP-GWC)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

1: Compute $\hat{\boldsymbol{\mu}}^{\text{obs}}, \hat{\boldsymbol{\sigma}}^{\text{obs}}$ using (14).
2: Compute $S_{\text{gwc}}^i \leftarrow \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ using $\hat{\Phi}^{\text{gwc}}$ defined in Lemma 4.
3: Compute $\hat{Q}_{1-\alpha}^{\text{gwc}} \leftarrow \hat{Q}_{1-\alpha}(S_{\text{gwc}}^1, \dots, S_{\text{gwc}}^n)$.
4: Compute $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \dots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$ using ω defined in Lemma 2.
5: **return** $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}}), \forall j \in [d]\right\}$.

Algorithm S6 Backward Search in Algorithm 2

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \dots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$; target miscoverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; mean index h^*; current coordinate j.

1: **for** $h \in \text{Row}_j(h^*)$ with $h_j = h_j^* - 1$ to 1 **do**
2: Compute $\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h))$ using Algorithm 1 and B_j^h using (24).
3: **if** $B_j^h > 0$ **then**
4: **return** $h^{[j]} = h$.
5: **end if**
6: **end for**

Algorithm S7 Binary Search in Algorithm 2

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \ldots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$; target miscoverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; mean index h^*; current coordinate j.

Let $I_{\min} \leftarrow h_j^*$ and $I_{\max} \leftarrow n+1$.
while $I_{\min} < I_{\max}$ **do**
  Compute $m_j \leftarrow \lceil (I_{\min} + I_{\max})/2 \rceil$ and $m_k = h_k^*$ for all $k \neq j$.
  Compute $\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^m))$ and B_j^m using (24). ▷ Algorithm 1
  if $B_j^m > 0$ **then**
    $I_{\min} \leftarrow m_j$.
  else
    $I_{\max} \leftarrow m_j - 1$.
  end if
  if $I_{\max} = I_{\min}$ **then**
    return $h_j^{[j]} = I_{\min}$ and $h_k^{[j]} = h_k^*$ for all $k \neq j$.
  end if
end while

A.3 Computational Cost Analysis

A.3.1 Algorithm S5

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n,d)-array, we know

1. Computing $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ using (14) requires $\mathcal{O}(nd)$.

2. Evaluating $S_{\text{gwc}}^i \leftarrow \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ for all $i \in [n]$ is also $\mathcal{O}(nd)$.

3. Computing the quantile $\hat{Q}_{1-\alpha}^{\text{gwc}}$ costs $\mathcal{O}(n)$ using a selection algorithm such as quickselect.

4. Computing $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \ldots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$ costs $\mathcal{O}(d)$ once $\hat{Q}_{1-\alpha}^{\text{gwc}}$ and $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ are known.

Hence, the overall computational complexity of Algorithm S5 is $\mathcal{O}(nd)$.

A.3.2 Algorithm 1

Algorithm 1 shares the same structure and computational cost as Algorithm S5: $\mathcal{O}(nd)$.

A.3.3 Algorithm 2

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n,d)-array, we know

1. Computing $(\hat{\mu}_j^{\text{obs}}, \hat{\sigma}_j^{\text{obs}})_{j=1}^d$ using (14) and $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \ldots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$ using Algorithm S5 requires $\mathcal{O}(nd)$.

2. Sorting $\{E_j^i\}_{i=1}^n$ for all $j \in [d]$ costs $\mathcal{O}(dn \log n)$;

3. Computing $\mathcal{Y}^{h^*}$ relies on computing h^*, which requires masking all values below $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and reporting the index of the lowest value from the sorted residuals list $\{E_j^i\}_{i=1}^n$, each of which costs $\mathcal{O}(n)$, so together this operation contributes $\mathcal{O}(dn)$.

4. If $\mathcal{Y}^{h^*} = \emptyset$, then the cost is only $\mathcal{O}(d)$ since all global bounds have been computed.

5. If $\mathcal{Y}^{h^*} \neq \emptyset$, then for each coordinate, we compute the local bounds using Algorithm 1 and $B_j^{h^*}$ using (24), which costs $\mathcal{O}(dn)$. If $B_j^{h^*} = 0$, then we use the backward search detailed in Algorithm S6, which costs $\mathcal{O}(dn^2)$; see Appendix A.3.4. If $B_j^{h^*} > 0$, then we use the binary search detailed in Algorithm S7, which costs $\mathcal{O}(dn \log n)$; see Appendix A.3.5.

Let $T \leq d$ be the number of coordinates where binary searches are enabled, then the total cost of Algorithm 2 is $\mathcal{O}(Tdn \log n + (d-T)dn^2)$.

A.3.4 Algorithm S6

Running Algorithm 1 and forming B_j^h costs $\mathcal{O}(dn)$ for each h; there are $n+1$ indices in $\text{Row}_j(h^*)$, so the computational complexity of Algorithm S6 never exceeds $\mathcal{O}(dn^2)$.

A.3.5 Algorithm S7

Running Algorithm 1 and forming B_j^h costs $\mathcal{O}(dn)$ for each h. The number of indices h encountered in this algorithm is $\mathcal{O}(\log_2 n)$ because we split and continue in only half of the current indices sequence at each step. This is a standard binary search. Therefore, the total computational complexity of Algorithm S7 never exceeds $\mathcal{O}(dn \log n)$.

Appendix B. Mathematical Proofs

B.1 Auxiliary Lemmas

Lemma S1 *Let $x^1, \ldots, x^n \in \mathbb{R}$ and $\bar{x}^n = \frac{1}{n}\sum_{i=1}^n x^i$. For any $x^{n+1} \in \mathbb{R}$, let $\hat{\sigma}(x^{n+1})$ be the sample standard deviation of $\{x^1, \ldots, x^n, x^{n+1}\}$, then*

$$\hat{\sigma}^2(x^{n+1}) = \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.$$

Proof Denote $\bar{x}^{n+1} = \frac{1}{n+1}\sum_{i=1}^{n+1} x^i$. We notice that

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\sum_{i=1}^{n+1}(x^i - \bar{x}^{n+1})^2 \\
=& \frac{1}{n}\left[(x^{n+1} - \bar{x}^{n+1})^2 + \sum_{i=1}^n (x^i - \bar{x}^n + \bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + \frac{1}{n}\sum_{i=1}^n \left[(x^i - \bar{x}^n)^2 + 2(x^i - \bar{x}^n)(\bar{x}^n - \bar{x}^{n+1}) + (\bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + (\bar{x}^n - \bar{x}^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

Substituting $\bar{x}^{n+1} = \frac{x^{n+1} + n\bar{x}^n}{n+1}$, we get

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\left(x^{n+1} - \frac{x^{n+1} + n\bar{x}^n}{n+1}\right)^2 + \left(\bar{x}^n - \frac{n\bar{x}^n + x^{n+1}}{n+1}\right)^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{n}{(n+1)^2}(x^{n+1} - \bar{x}^n)^2 + \frac{1}{(n+1)^2}(\bar{x}^n - x^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

■

Lemma S2 *Let $X^1, \ldots, X^n \in \mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^n$. Let X^{n+1} and $\tilde{X}^{n+1}$ be another two random variables such that*

$$|X^{n+1} - \bar{X}^n| \leq |\tilde{X}^{n+1} - \bar{X}^n| \quad a.s.$$

Then,

$$\hat{\sigma}(X^{n+1}) \leq \hat{\sigma}(\tilde{X}^{n+1}) \quad a.s.,$$

where $\hat{\sigma}(X^{n+1})$ is the sample standard deviation of $\{X^1, \ldots, X^n, X^{n+1}\}$, as in Lemma S1.

Proof This is a direct application of Lemma S1:

$$\begin{aligned}
\hat{\sigma}^2(X^{n+1}) &= \frac{(X^{n+1}-\bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^{n}(X^i-\bar{X}^n)^2 \\
&\leq \frac{(\tilde{X}^{n+1}-\bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^{n}(X^i-\bar{X}^n)^2 \quad \text{[Assumption]} \\
&= \hat{\sigma}^2(\tilde{X}^{n+1}),
\end{aligned}$$

and the claim follows by taking square roots. ■

Lemma S3 *Let $X^1,\dots,X^{n+1} \in \mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^{n+1}$ and sample standard deviation $\bar{S}^{n+1} > 0$. Define $Z^i = (X^i - \bar{X}^{n+1})/\bar{S}^{n+1}$ for $i \in [n+1]$. Then,*

$$|Z^i| \leq \frac{n}{\sqrt{n+1}} \quad \textit{almost surely}, \quad \forall i \in [n+1].$$

This inequality is strict if $X^1,\dots,X^{n+1}$ are almost-surely distinct.

Proof Fix any $i \in [n+1]$. Let $\bar{X}^{-i} = \frac{1}{n}\sum_{k\neq i} X^k$ be the sample mean of the random variables excluding the i-th one, and

$$\bar{S}^{-i} = \sqrt{\frac{1}{n}\sum_{k\neq i}(X^k - \bar{X}^{-i})^2}.$$

Then, it follows from Lemma S1 that

$$|Z^i| = \frac{|X^i-\bar{X}^{n+1}|}{\bar{S}^{n+1}} = \frac{\left|X^i - \frac{1}{n+1}\sum_{k=1}^{n+1}X^k\right|}{\sqrt{\frac{1}{n}\sum_{k=1}^{n+1}(X^k-\bar{X}^{n+1})^2}} = \frac{\frac{n}{n+1}|X^i-\bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i-\bar{X}^{-i})^2+(\bar{S}^{-i})^2}}.$$

But $\bar{S}^{-i} \geq 0$, so we must have

$$|Z^i| \leq \frac{\frac{n}{n+1}|X^i-\bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i-\bar{X}^{-i})^2}} = \frac{n}{\sqrt{n+1}}.$$

If $X^1,\dots,X^{n+1}$ are almost-surely distinct, we know $\bar{S}^{-i} > 0$ almost-surely for any $i \in [n+1]$ thus this inequality becomes strict. ■

B.2 Proofs of Main Results

Proof [of Theorem 1] This is a well-known result, whose proof is included for completeness. By definition of $\hat{C}(\mathbf{X}^{n+1})$ in (5),

$$\mathbf{Y}^{n+1} \notin \hat{C}(\mathbf{X}^{n+1}) \iff S^{n+1} > \hat{Q}_{1-\alpha}(S^1,\dots,S^n).$$

Moreover, $S^1, \ldots, S^{n+1}$ are exchangeable because $\mathbf{E}^1, \ldots, \mathbf{E}^{n+1}$ are exchangeable and Φ is fixed. Therefore, the proof is complete by applying the fundamental "quantile inflation lemma" of conformal prediction (e.g., see Angelopoulos et al. (2024)), which states

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \leq \alpha.$$

Moreover, it is also a standard result that, if $S^1, \ldots, S^{n+1}$ are almost-surely distinct,

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \geq \alpha - \frac{1}{n+1}.$$

■

Proof [of Lemma 2] From Assumption 1, we know that $E_j^1, \ldots, E_j^{n+1}$ are almost-surely distinct for every $j \in [d]$ and $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost-surely for all $j \in [d]$, which also ensures S_{oracle}^{n+1} is well-defined. To prove this lemma, fix any $c \in \mathbb{R}$ and note that

$$\begin{aligned}
&S_{\text{oracle}}^{n+1} \leq c \\
&\iff E_j^{n+1} \leq c\,\hat{\sigma}_j^{\text{oracle}} + \hat{\mu}_j^{\text{oracle}}, \quad \forall j \in [d] \\
&\iff E_j^{n+1} - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i,\right)^2} - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i \leq 0, \quad \forall j \in [d] \\
&\iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i,\right)^2} \leq 0, \quad \forall j \in [d] \\
&\iff \underbrace{\frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2}}_{=:g_j(\mathbf{E}^{n+1})} \leq 0, \quad \forall j \in [d]
\end{aligned}$$

where the last implication follows from Lemma S1. Everything except $\mathbf{E}^{n+1}$ here is observed, so we can solve this inequality for $\mathbf{E}^{n+1}$ coordinate-wise.

Fix any $j \in [d]$. It follows from the above that

$$g_j(\mathbf{E}^{n+1}) \leq 0 \iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) \leq c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2}. \tag{S1}$$

The solution of this inequality in E_j^{n+1} depends on the range of c. There are two cases:

1. if $c \geq 0$, then (S1) holds if and only if *either* $0 \leq E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *or* $E_j^{n+1} \geq \hat{\mu}_j^{\text{obs}}$ and

$$\left(\frac{n}{n+1}\right)^2\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \leq c^2\left(\frac{1}{n+1}(E_j^{n+1} - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

2. if $c < 0$, then (S1) holds if and only if $E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *and*

$$\left(\frac{n}{n+1}\right)^2\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \geq c^2\left(\tfrac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

The solution in both cases relies on solving a quadratic equality in the form $h(x) = 0$, where:

$$h(x) := \left(\frac{n}{n+1}\right)^2 \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - c^2 \tfrac{1}{n+1} \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - c^2 (\hat{\sigma}_j^{\text{obs}})^2 = 0.$$

This equality can be reorganized into a more recognized quadratic form

$$h(x) = Ax^2 + Bx + K = 0,$$

where

$$\begin{aligned}
A &= \left(\frac{n}{n+1}\right)^2 - \frac{c^2}{n+1}, \\
B &= 2\hat{\mu}_j^{\text{obs}} \left(\frac{n}{n+1}\right)^2 - 2\hat{\mu}_j^{\text{obs}} \frac{c^2}{n+1}, \\
K &= \left(\frac{n}{n+1}\right)^2 (\hat{\mu}_j^{\text{obs}})^2 - \frac{c^2}{n+1} (\hat{\mu}_j^{\text{obs}})^2 - c^2 (\hat{\sigma}_j^{\text{obs}})^2.
\end{aligned}$$

The solution set is:

$$h(x) = 0 \iff x = \begin{cases} \hat{\mu}_j^{\text{obs}} \pm \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c^2 < \frac{n^2}{n+1}, \\ \hat{\mu}_j^{\text{obs}}, & c^2 = \frac{n^2}{n+1} \text{ and } \hat{\sigma}_j^{\text{obs}} = 0, \\ \text{no real solution}, & \text{otherwise}. \end{cases}$$

We first note the event $\hat{\sigma}_j^{\text{obs}} = 0$ occurs if and only if $E_j^1 = \ldots = E_j^n$, which has probability zero by Assumption 1. For the rest, we note that $c^2 < \frac{n^2}{n+1}$ implies $A > 0$ thus $h(x) = Ax^2 + Bx + K$ concave up for all $x \geq 0$. Therefore by analyzing the positiveness and negativeness of $h(x)$ relative to the solution set in our two cases, we know (S1) holds almost-surely if and only if

$$\begin{cases} \hat{\mu}_j^{\text{obs}} \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c \geq 0,\ c^2 < \frac{n^2}{n+1}, \\ E_j^{n+1} < \infty, & c \geq 0,\ c^2 \geq \frac{n^2}{n+1}, \\ 0 \leq E_j^{n+1} < \hat{\mu}_j^{\text{obs}}, & c \geq 0, \\ 0 \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c < 0,\ c^2 < \frac{n^2}{n+1}, \\ \text{no solution}, & c < 0,\ c^2 \geq \frac{n^2}{n+1}. \end{cases} \quad \text{almost-surely}$$

$$\iff \begin{cases} 0 \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & 0 \leq c < \frac{n}{\sqrt{n+1}}, \\ 0 \leq E_j^{n+1} < \infty, & c \geq \frac{n}{\sqrt{n+1}}, \\ 0 \leq E_j^{n+1} \leq \max\left\{0, \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}\right\}, & -\frac{n}{\sqrt{n+1}} < c < 0, \\ \text{no solution}, & c \leq -\frac{n}{\sqrt{n+1}}. \end{cases} \quad \text{almost surely}$$

Interpreting the "no-solution" case as $E_j^{n+1} = 0$, we obtain precisely the expression for $\omega_j(c)$ given in the statement of the lemma. This notation adjustment is harmless because both $E_j^{n+1} = 0$ and $S_{\text{oracle}}^{n+1} \le -n/\sqrt{n+1}$ are probability null sets by our Assumption 1 and Lemma S3, respectively. The monotonic nondecreasing property of this $\omega_j(c)$ follows immediately from the construction. ■

Proof [of Lemma 3] From Assumption 1, $E_j^1, \dots, E_j^{n+1}$ are almost-surely distinct for every $j \in [d]$ and $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost-surely for all $j \in [d]$. This ensures S_{oracle}^i is well-defined for all $i \in [n+1]$. Moreover, we know from Lemma S3 that

$$\left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \frac{n}{\sqrt{n+1}},$$

almost-surely for all $i \in [n+1]$. Aggregating this inequality for all $j \in [d]$, we obtain

$$|S_{\text{oracle}}^i| = \left| \max_{1 \le j \le d} \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| \le \max_{1 \le j \le d} \left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \max_{1 \le j \le d} \frac{n}{\sqrt{n+1}} = \frac{n}{\sqrt{n+1}},$$

almost-surely for all $i \in [n+1]$. Recall that the quantile $\hat{Q}_{1-\alpha}^{\text{oracle}}$ is defined as

$$\hat{Q}_{1-\alpha}^{\text{oracle}} := \lceil (1-\alpha)(n+1) \rceil \text{th smallest value in } \{S_{\text{oracle}}^1, \dots, S_{\text{oracle}}^n, +\infty\},$$

so $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost-surely if $n \ge 1/\alpha - 1$ and $\hat{Q}_{1-\alpha}^{\text{oracle}} = +\infty$ otherwise. ■

Proof [of Lemma 4] Fix any $j \in [d]$ and let

$$g(t_j, z_j) := \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}.$$

Finding the supremum of this g over $z_j \ge 0$ is a simple univariate optimization problem. Expanding g in full, we get

$$g(t_j, z_j) = \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}},$$

where the last equality follows from Lemma S1. Differentiating g with respect to z_j yields

$$g_{z_j}(t_j, z_j) = \frac{-\frac{1}{n+1}\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2} - \left(t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})\right)\frac{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}}}{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}.$$

Solving $g_{z_j}(t_j, z_j^*) = 0$ with respect to z_j^* gives:

$$(\hat{\sigma}_j^{\text{obs}})^2 + (t_j - \hat{\mu}_j^{\text{obs}})(z_j^* - \hat{\mu}_j^{\text{obs}}) = 0 \quad \Longrightarrow \quad z_j^* = \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}.$$

There are three cases here for any fixed $t_j \ge 0$:

- Case 1: if $g(t_j, z_j^*)$ is indeed the global maximum and $z_j^* > 0$, then we can just take

$$\sup_{z_j \geq 0} g(t_j, z_j) = g(t_j, z_j^*).$$

- Case 2: if $g(t_j, z_j^*)$ is not the global maximum or $z_j^* < 0$, then the supremum must occur at either $z_j = 0$ or $z_j \to \infty$ due to the fact that g_{z_j} has only one critical point. Hence,

$$\sup_{z_j \geq 0} g(t_j, z_j) = \max\left\{ g(t_j, 0), \lim_{z_j \to \infty} g(t_j, z_j) \right\} = \max\left\{ g(t_j, 0), -\frac{1}{\sqrt{n+1}} \right\}.$$

- Case 3: if $t_j = \hat{\mu}_j^{\text{obs}}$, then the same derivation shows that the numerator of g_{z_j} reduces to $-\frac{1}{n+1}(\hat{\sigma}_j^{\text{obs}})^2 < 0$. This implies that g is monotonically decreasing and thus we take

$$\sup_{z_j \geq 0} g(\hat{\mu}_j^{\text{obs}}, z_j) = g(\hat{\mu}_j^{\text{obs}}, 0).$$

Collecting all candidates, we obtain the closed form:

$$\sup_{z_j \geq 0} g(t_j, z_j) = \max\left\{ g\left(t_j, 0\right), g\left(t_j, z_j^* \mathbb{I}\left\{z_j^* \geq 0\right\}\right), -\frac{1}{\sqrt{n+1}} \right\}.$$

■

Proof [of Proposition 6] Throughout this proof we write $\tilde{C}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ for the estimated oracle prediction set defined in (11). Let $\mathcal{P}$ be any (random) partition of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$, so that $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) = \bigsqcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R}$ almost-surely. Because $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \supseteq \tilde{C}(\mathbf{X}^{n+1})$ almost-surely, we know

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\iff \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \mathcal{R} \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R}.
\end{aligned}$$

If $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ satisfies the containment on the right-hand side of (19),

$$\mathcal{R} \supseteq \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \supseteq \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R},$$

then

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\implies \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\iff \mathbf{Y}^{n+1} \in \bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\implies \mathbf{Y}^{n+1} \in \text{Rect}_{\mathcal{P}}\left(\bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \right) = \hat{C}_{\mathcal{P}}^{\text{lwc}}(\mathbf{X}^{n+1}),
\end{aligned}$$

where the last implication holds because $\mathrm{Rect}_{\mathcal{P}}(A) \supseteq A$ by definition. Since we already know that $\mathbb{P}[\mathbf{Y}^{n+1} \in \widetilde{C}(\mathbf{X}^{n+1})] \geq 1-\alpha$, by Theorem 1 and the dual relation (12), this proves that

$$\mathbb{P}\left[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})\right] \geq 1-\alpha.$$

Finally, the containment $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ almost-surely is immediate from the definition of $\mathrm{Rect}_{\mathcal{P}}$ in (20), because

$$\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \subseteq \mathcal{R} \subseteq \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1}),$$

for every $\mathcal{R} \in \mathcal{P}$. ■

Proof [of Lemma 7] Fix any $h \in [n+1]^d$. Suppose $\mathcal{Y}^h \neq \emptyset$, then $\mathcal{E}^h \neq \emptyset$ and, by Equation (21),

$$\mathcal{E}^h = \bigtimes_{j=1}^{d} \mathcal{E}^h_j, \quad \text{where} \quad \mathcal{E}^h_j := \left[E_j^{(h_j-1)}, U_j^{h_j}\right) \neq \emptyset, \quad \forall j \in [d].$$

Fix any $\mathbf{t} \in \mathbb{R}^d_+$. Similar to the proof of Lemma 4, we have to solve a univariate optimization problem. We first notice that for any $j \in [d]$,

$$\sup_{\mathbf{z} \in \mathcal{E}^h} \frac{t_j}{\hat{\sigma}_j(z_j)} = \sup_{z_j \in \mathcal{E}^h_j} \frac{t_j}{\hat{\sigma}_j(z_j)} = \frac{t_j}{\inf_{z_j \in \mathcal{E}^h_j} \hat{\sigma}_j(z_j)}.$$

By Lemma S1, we know that

$$\hat{\sigma}_j(z_j) = \sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2},$$

which is continuous in z_j, so the infimum above is attained on the closure $\overline{\mathcal{E}^h_j} = \left[E_j^{(h_j-1)}, U_j^{h_j}\right]$; clearly $\hat{\sigma}_j(z_j)$ is minimized there at

$$z_j^* = \underset{z_j \in \overline{\mathcal{E}^h_j}}{\mathrm{argmin}}\, |z_j - \hat{\mu}_j^{\mathrm{obs}}| = \begin{cases} \hat{\mu}_j^{\mathrm{obs}}, & \text{if } \hat{\mu}_j^{\mathrm{obs}} \in \overline{\mathcal{E}^h_j}, \\ E_j^{(h_j-1)}, & \text{if } \hat{\mu}_j^{\mathrm{obs}} < E_j^{(h_j-1)}, \\ U_j^{h_j}, & \text{if } \hat{\mu}_j^{\mathrm{obs}} > U_j^{h_j}. \end{cases}$$

Hence, $r_j(h_j) = \hat{\sigma}_j(z_j^*)$ is exactly the infimum proposed. For the second part, denote

$$g(z_j) := \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{\frac{1}{n+1}(z_j + n\hat{\mu}_j^{\mathrm{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2}} = \frac{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}}) + \hat{\mu}_j^{\mathrm{obs}}}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2}},$$

then we want to find

$$\inf_{\mathbf{z} \in \mathcal{E}^{\mathrm{gwc}}} g(z_j) = \inf_{z_j \in [0, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})]} g(z_j).$$

Differentiating $g(z_j)$ in z_j, we get

$$g'(z_j) = \frac{\frac{1}{n+1}(\hat{\sigma}_j^{\mathrm{obs}})^2 - \frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})\hat{\mu}_j^{\mathrm{obs}}}{(\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2)\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2}}.$$

Setting this to zero gives the critical point

$$z_j^* = \hat{\mu}_j^{\text{obs}} + \frac{(\hat{\sigma}_j^{\text{obs}})^2}{\hat{\mu}_j^{\text{obs}}} > 0.$$

Let $\epsilon > 0$, then we see that

$$\begin{aligned}
\text{sign}(g'(z_j^* + \epsilon)) &= \text{sign}\left(-\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = -1, \\
\text{sign}(g'(z_j^* - \epsilon)) &= \text{sign}\left(\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = +1.
\end{aligned}$$

This implies that $g(z_j^*)$ is always a local maximum. Hence, if $z_j^* \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]$, then

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = \min\left\{ g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} g(z_j) \right\};$$

if $z_j^* > \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})$, then g is nondecreasing on $[0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]$ and

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = g(0).$$

But in either case, we only need to compare the boundaries in order to find the infimum. Therefore, it follows that

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = \min\left\{ g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} g(z_j) \right\}.$$

This is true for every $j \in [d]$, so we must have

$$\begin{aligned}
\hat{\Phi}_h^{\text{lwc}}(\mathbf{t}) &= \max_{1 \le j \le d} \left\{ \sup_{\mathbf{z} \in \mathcal{E}^h} \frac{t_j}{\hat{\sigma}_j(z_j)} - \inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} g(z_j) \right\} \\
&= \max_{1 \le j \le d} \left\{ \frac{t_j}{r_j(h_j)} - \min\left\{ \frac{\hat{\mu}_j(0)}{\hat{\sigma}_j(0)}, \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \right\} \right\}.
\end{aligned}$$

■

Proof [of Lemma 8] Fix any $j \in [d]$ and suppose $\mathcal{Y}^{h^*} \neq \emptyset$. There are two parts to this lemma, which we prove separately. Throughout, for any $k \in [n+1]$ we let $h^*(j,k)$ denote the unique multi-index in $\text{Row}_j(h^*)$ whose j-th coordinate equals k; that is, $h_j^*(j,k) = k$ and $h_l^*(j,k) = h_l^*$ for all $l \neq j$.

Part I. We aim to prove

$$\sup_{h \in [n+1]^d} B_j^h = \sup_{h \in \text{Row}_j(h^*)} B_j^h = B_j^{h^{[j]}}. \tag{S2}$$

We first notice that the first supremum can be decomposed into suprema over surfaces:

$$\sup_{h \in [n+1]^d} B_j^h = \sup_{k \in [n+1]} \sup_{h \in \text{Surface}_j(k)} B_j^h,$$

where

$$\text{Surface}_j(k) := \left\{ h \in [n+1]^d : h_j = k \right\}.$$

Fix any $k \in [n+1]$ and note that $h^*(j,k) \in \text{Surface}_j(k) \cap \text{Row}_j(h^*)$. Since this holds for every $k \in [n+1]$, to prove (S2) the main challenge is to show that:

$$\sup_{h \in \text{Surface}_j(k)} B_j^h = B_j^{h^*(j,k)}. \tag{S3}$$

To prove (S3), recall from the definition in (22) and Lemma 7 that

$$\hat{\Phi}_h^{\text{lwc}}(\mathbf{t}) = \max_{1 \le l \le d} \left\{ \frac{t_l}{r_l(h_l)} - c_l \right\},$$

where

$$r_l(h_l) = \min\left\{ \hat{\sigma}_l(z_l) : z_l \in \overline{\mathcal{E}_l^h} \right\}, \quad \overline{\mathcal{E}_l^h} := \left[E_l^{(h_l-1)}, U_l^{h_l} \right], \qquad c_l = \inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} \frac{\hat{\mu}_l(z_l)}{\hat{\sigma}_l(z_l)}.$$

Without loss of generality, assume $\mathcal{Y}^h \neq \emptyset$ for all $h \in \text{Surface}_j(k)$. This does not affect the final result because $\mathcal{Y}^h = \emptyset$ leads to $B_j^h = 0$, which can be safely excluded from the supremum calculation. Then notice that, for all $h \in \text{Surface}_j(k)$, we have $r_j(h_j) = r_j(k)$ and $r_l(h_l) \ge \hat{\sigma}_l^{\text{obs}} = r_l(h_l^*)$ for all $l \neq j$, the latter because $\hat{\mu}_l^{\text{obs}} \in \overline{\mathcal{E}_l^{h^*}}$ by the definition of the mean index h^*. Therefore, for any fixed $\mathbf{t}$,

$$\hat{\Phi}_h^{\text{lwc}}(\mathbf{t}) = \max_{1 \le l \le d} \left\{ \frac{t_l}{r_l(h_l)} - c_l \right\} \le \max\left(\max_{l \neq j} \left\{ \frac{t_l}{\hat{\sigma}_l^{\text{obs}}} - c_l \right\}, \frac{t_j}{r_j(k)} - c_j \right) = \hat{\Phi}_{h^*(j,k)}^{\text{lwc}}(\mathbf{t}).$$

Because this holds point-wise for all $\mathbf{t}$, we can extend the inequality to the corresponding conformal quantile and conclude that, almost surely for all $h \in \text{Surface}_j(k)$,

$$\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h) \le \hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})$$

and therefore, by the monotonicity of the function ω_j from Lemma 2,

$$\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h)) \le \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})).$$

Therefore, it follows that, for all $h \in \text{Surface}_j(k)$,

$$\begin{aligned}
B_j^h &= \begin{cases} \min\left\{ U_j^k, \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h)) \right\}, & \text{if } U_j^k,\ \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h)) > E_j^{(k-1)}, \\ 0, & \text{otherwise}, \end{cases} \\
&\le \begin{cases} \min\{ U_j^k, \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})) \}, & \text{if } U_j^k,\ \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})) > E_j^{(k-1)}, \\ 0, & \text{otherwise}, \end{cases} \\
&= B_j^{h^*(j,k)}.
\end{aligned}$$

This completes the proof of (S3).

The second equality in (S2) is trivial. We can easily see that for any $k \in [n]$, if $B_j^{h^*(j,k+1)} \neq 0$, then

$$U_j^{k+1},\ \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k+1)})) \geq E_j^{(k)} \geq \min\left\{E_j^{(k)}, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})\right\} = U_j^k.$$

Hence, it follows that $B_j^{h^*(j,k+1)} \geq B_j^{h^*(j,k)}$ regardless of whether $B_j^{h^*(j,k)} = 0$ or not. Then finding the supremum among all $k \in [n+1]$ simplifies to finding the right-most nonzero $B_j^{h^*(j,k)}$, which is exactly how $B_j^{h^{[j]}}$ is defined.

Part II. In this part, we aim to show that if $B_j^{h^*(j,m_j)} = 0$ for some $m_j \geq h_j^*$, then

$$B_j^{h^*(j,N)} = 0, \quad \forall N \geq m_j.$$

There are two cases that could lead to $B_j^{h^*(j,m_j)} = 0$:

- If $U_j^{m_j} = \min\left\{E_j^{(m_j)}, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})\right\} \leq E_j^{(m_j-1)}$, then $\omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}}) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)}$ for any $N \geq m_j$ by the nature of order statistics. Hence, $U_j^N \leq E_j^{(N-1)}$ for all $N \geq m_j$, which further implies $B_j^{h^*(j,N)} = 0$ for all $N \geq m_j$.

- If $\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)})) \leq E_j^{(m_j-1)}$, then we claim that

$$\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,N)})) \leq \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)})) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)},$$

  which then implies $B_j^{h^*(j,N)} = 0$ for all $N \geq m_j$. This is actually trivial because $m_j \geq h_j^*$ implies that both $\overline{\mathcal{E}_j^{h^*(j,m_j)}}$ and $\overline{\mathcal{E}_j^{h^*(j,N)}}$ lie to the right of $\hat{\mu}_j^{\text{obs}}$, with the latter no closer to it than the former, so that

$$r_j(m_j) = \min_{z_j \in \overline{\mathcal{E}_j^{h^*(j,m_j)}}} \hat{\sigma}_j(z_j) \leq \min_{z_j \in \overline{\mathcal{E}_j^{h^*(j,N)}}} \hat{\sigma}_j(z_j) = r_j(N),$$

  by Lemma S2 and the definition of $\mathcal{E}_j^h$. This then leads to

$$\begin{aligned}
&\hat{\Phi}_{h^*(j,m_j)}^{\text{lwc}}(\mathbf{t}) \geq \hat{\Phi}_{h^*(j,N)}^{\text{lwc}}(\mathbf{t}), \quad \forall \mathbf{t} \\
&\implies \hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,N)}) \leq \hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)}) \\
&\implies \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,N)})) \leq \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)})).
\end{aligned}$$

■

Appendix C. Additional Numerical Results

This appendix reports the sensitivity analyses and stress tests supporting Section 4. Unless stated otherwise, the absolute-residual experiments use $d = 10$, target coverage 0.9, and 200 repetitions. The regression and evaluation protocol is the same as in Section 4.2: ordinary least squares is fit on 7,200 independently drawn training observations, and coverage is evaluated on 800 fresh test observations. The calibration sample is also independently redrawn in every repetition, while the underlying regression coefficients are held fixed. The CQR experiments use 30 repetitions with 2,400 training and 600 test observations per repetition; the shape-template comparison uses the same training/test sizes and 100 repetitions. Figures show mean curves without error bars, while the accompanying CSV files retain across-repetition standard deviations.

C.1 Coverage Uncertainty Audit

We first audited 58 TSCP settings in the synthetic figures and the retained real-data table. This initial audit concerns those figures and the retained main-text real-data table; it does not audit the historical figures in Appendix D. The corresponding nominal target lies within one across-repetition standard deviation of every reported mean. Only four synthetic means are below their targets: partial heteroskedasticity at $n = 500$ gives 0.8996 (0.0164); capped-score CQR at $n = 50$ gives 0.8991 (0.0457); 5% contamination gives 0.8995 (0.0297); and Student $t_{1.5}$ noise at $n = 500$ gives 0.8976 (0.0173). Parentheses in this paragraph contain one standard deviation across repetitions.

For the revised synthetic experiments, training, test, and calibration observations are independently redrawn, so the repetitions are independent conditional on the fixed regression coefficients. This interpretation does not turn repeated splits of a fixed real dataset into fresh population draws. Dividing the preceding standard deviations by the square root of the number of repetitions therefore gives a Monte Carlo standard error. The resulting normal-approximation 95% intervals are, respectively, $[0.89734, 0.90189]$, $[0.88271, 0.91541]$, $[0.89536, 0.90360]$, and $[0.89522, 0.90003]$; each contains the target 0.9. Thus, the small displayed shortfalls are consistent with finite Monte Carlo averaging and do not reveal a systematic coverage defect in these experiments. The complete pointwise check is provided in `coverage_uncertainty_audit.csv` and can be checked from the accompanying trial-level CSV files in `experiment_data`. These pointwise normal approximations are not simultaneous confidence intervals or proofs of coverage. The separate follow-up real-data cohort is reported in Appendix C.9; its uncertainty summaries are not included in this original count of 58 settings.

C.2 Dependence Strength

We first vary the equicorrelation of the Gaussian noise over $\rho \in \{0, 0.3, 0.6, 0.9\}$ while retaining marginal standard deviations $(10, 9, \ldots, 1)$ and a calibration size of $n = 100$. Figure S1 shows that TSCP's empirical joint coverage is stable across this range: 0.905, 0.905, 0.907, and 0.908, respectively. Its average residual-space volume decreases from 6.37×10^{10} at $\rho = 0$ to 9.47×10^{9} at $\rho = 0.9$. Point CHR and Unscaled Max also remain close to the target. Emp. Copula improves as the correlation strengthens but remains below the nominal level

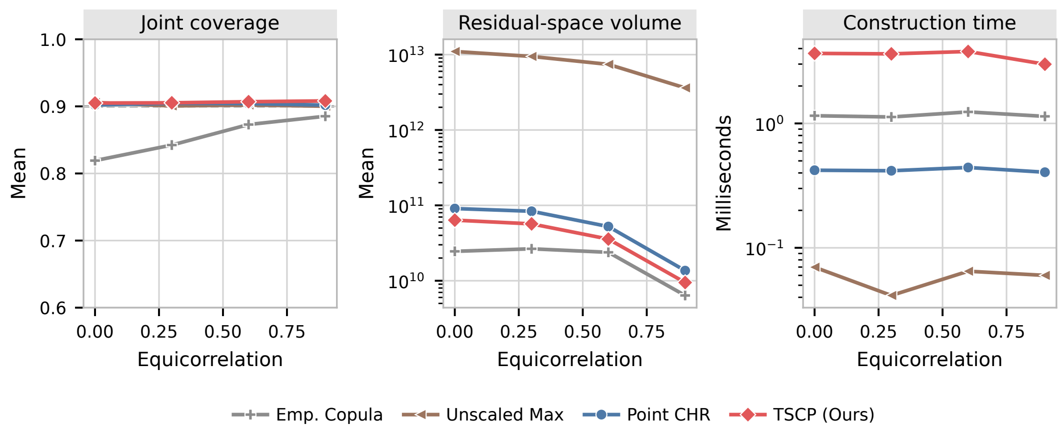


Figure S1: Sensitivity to equicorrelated Gaussian noise at $d = 10$ and $n = 100$. The three panels report joint coverage, mean residual-space volume, and mean construction time. Results average 200 repetitions; the dashed line marks the target coverage.

throughout the sweep. The experiment therefore does not indicate a coverage failure of TSCP under strong equicorrelation, while also showing that dependence materially changes region size.

C.3 Severity of Coordinate Heterogeneity

To isolate heterogeneity across outcomes, we use independent Gaussian noise and choose ten coordinate scales geometrically spaced between r and 1, where $r \in \{1, 2, 5, 10, 20\}$. The calibration size is fixed at $n = 100$. Figure S2 shows that the coverage of TSCP, Point CHR, and Unscaled Max is essentially unchanged as r grows. At $r = 1$, Unscaled Max is slightly smaller than TSCP, with mean volumes 1.43×10^4 and 1.80×10^4. At $r = 20$, their mean volumes are 3.44×10^{15} and 5.75×10^{10}, respectively. This widening gap is the expected benefit of coordinate adaptation: a single maximum-residual threshold becomes increasingly inefficient when one scale must be used for all coordinates. Emp. Copula remains smaller but has average coverage about 0.821 in this experiment.

C.4 Target Level and Small Calibration Samples

Figure S3 repeats the dependent Gaussian experiment at nominal coverage levels 0.8, 0.9, and 0.95, with $n = 100$ and $\rho = 0.6$. TSCP achieves empirical coverages 0.811, 0.907, and 0.954. As expected, increasing the target level increases all region volumes. The relative ordering is stable: TSCP is smaller than Point CHR and dramatically smaller than Unscaled Max, while Emp. Copula is smaller but below the nominal coverage at every displayed target.

We separately examine $n \in \{10, 20, 30, 50, 100\}$ under independent Gaussian noise. Figure S4 distinguishes finite volumes from the frequency of infinite output. At $n = 10$, Point CHR has coverage one and infinite volume in every repetition. This is a finite-sample quantile effect, not evidence of a superior region: Point CHR divides the calibration sample and its final subset is too small to supply a finite 0.9 conformal quantile. Point CHR is finite in all repetitions once $n \geq 20$. TSCP and Unscaled Max are finite throughout; both are

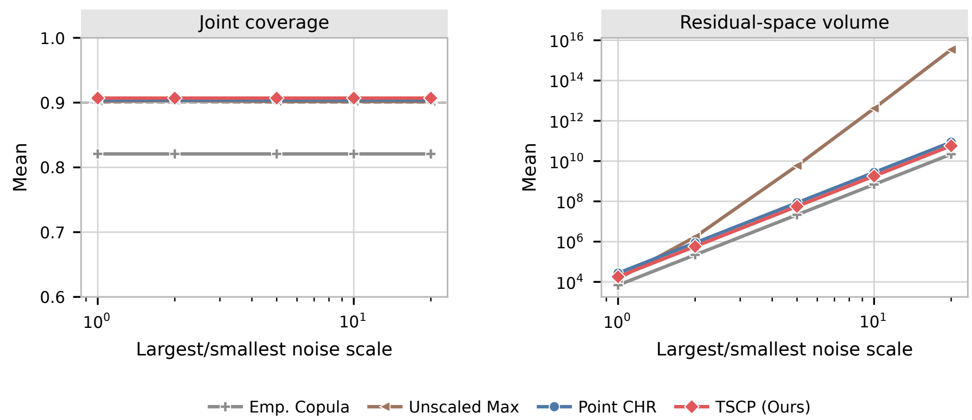


Figure S2: Sensitivity to the ratio between the largest and smallest outcome noise scales, with independent Gaussian errors and $n = 100$. The scales are geometrically spaced, and both axes of the volume panel are logarithmic. Results average 200 repetitions.

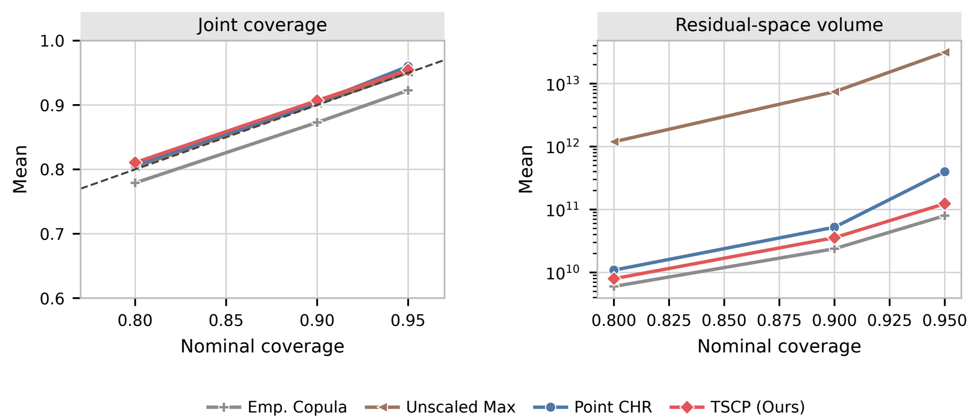


Figure S3: Sensitivity to the nominal coverage level in the $d = 10$ equicorrelated Gaussian experiment with $n = 100$ and $\rho = 0.6$. The dashed diagonal in the left panel is exact nominal calibration. Results average 200 repetitions.

conservative at the smallest sample sizes, as expected from the discrete conformal rank. Emp. Copula severely undercovers when n is small because it estimates dependence and calibrates on the same limited sample.

C.5 Heavy-Tailed Noise and the Role of Moment Conditions

The original heavy-tail experiment is reorganized in Figure S5. It uses independent Student t noise with $d = 10$, a common unit scale across coordinates, and degrees of freedom $\nu \in \{1.5, 2, 3\}$. The cases $\nu = 1.5$ and $\nu = 2$ have infinite variance and therefore lie outside Assumption 1, which is invoked by the validity theorem as well as the moment-based motivation; $\nu = 3$ has finite variance but remains heavy-tailed. These results should be

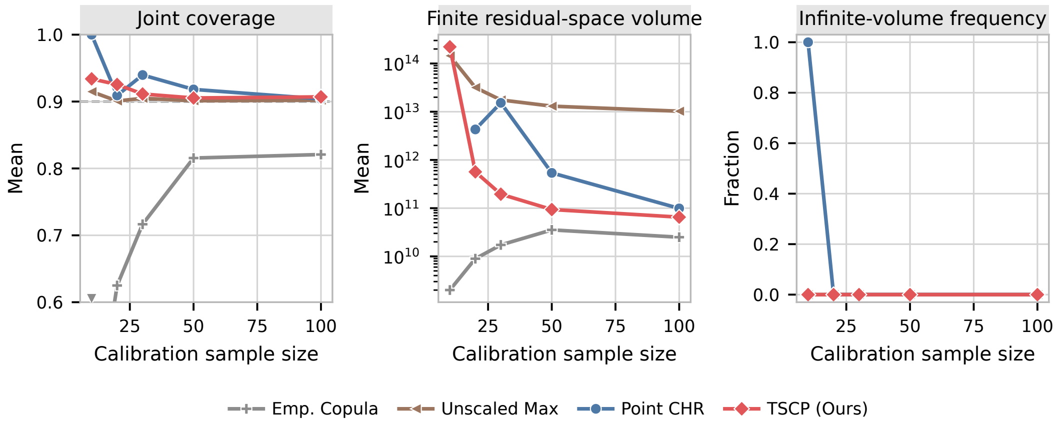


Figure S4: Small-calibration-sample stress test with independent Gaussian noise and scales $(10, 9, \ldots, 1)$. The middle panel averages only finite residual-space volumes; the right panel reports the fraction of repetitions with infinite volume. A downward triangle at the lower plotting boundary marks the Emp. Copula mean below 0.6. Results average 200 repetitions.

read as an empirical stress test, not as an extension of the stated theory to infinite-variance distributions.

TSCP remains near the target in the displayed cases. For example, at $n = 30$ its coverages are 0.909, 0.908, and 0.912 as ν increases, while at $n = 500$ they are 0.898, 0.901, and 0.900. Efficiency is much less stable. At $n = 30$, the mean TSCP volume falls from 1.59×10^{20} for $\nu = 1.5$ to 5.92×10^{13} for $\nu = 3$; at $n = 500$, it falls from 8.42×10^{13} to 6.07×10^{7}. The explosive volumes make clear that empirical coverage alone is not evidence of practically useful behavior under very heavy tails.

C.6 Exchangeable Contamination Stress Test

To test sensitivity to atypically large residuals, we contaminate the Gaussian experiment at rates $\varepsilon \in \{0, 0.01, 0.05, 0.10\}$. Independently for each observation, all ten noise coordinates are multiplied by 10 with probability ε; the same exchangeable mechanism is used for training, calibration, and test data. The calibration size is $n = 100$. We report median rather than mean volume because the contaminated distribution produces a small number of extreme regions.

Figure S6 shows that TSCP coverage changes from 0.904 without contamination to 0.902 at 10% contamination, but its median volume grows from 5.38×10^{10} to 1.10×10^{17}. Point CHR and Unscaled Max exhibit the same qualitative inflation, and Emp. Copula under-covers at every contamination level. Thus, the conformal calibration is comparatively stable in coverage, whereas the sample mean and standard deviation used for coordinate scaling are not robust efficiency summaries. Replacing them by robust estimators would change the construction and its analysis, so we leave that variant for future work rather than implying a guarantee not established here.

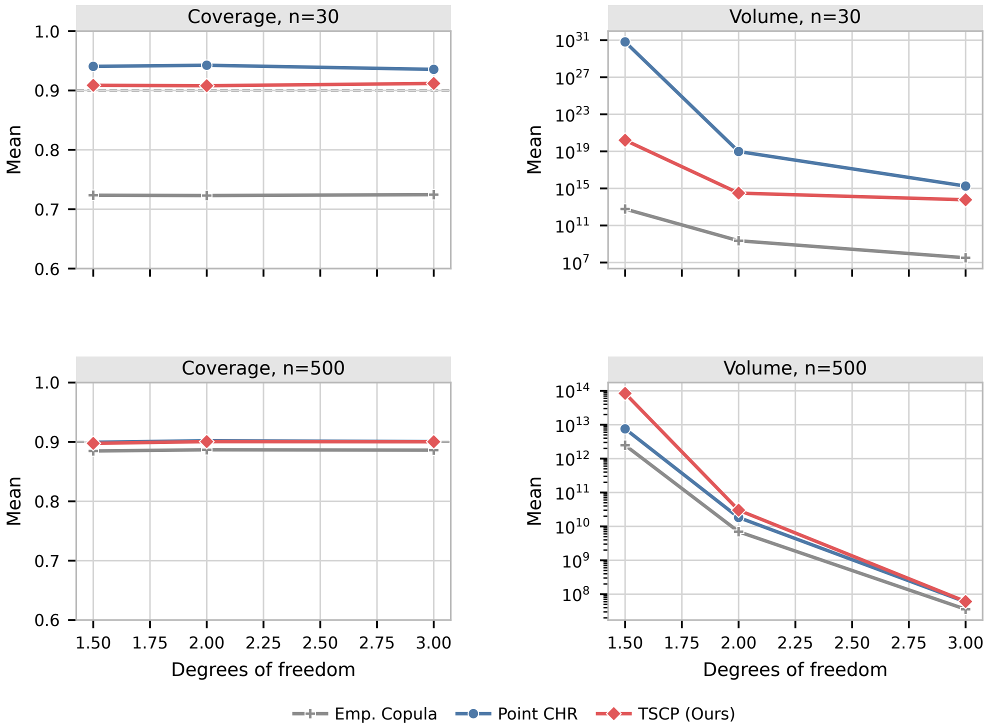


Figure S5: Coverage and mean residual-space volume under independent, common-scale Student t noise for $d = 10$. The top row uses $n = 30$ calibration examples and the bottom row uses $n = 500$. Results average 200 repetitions; volume is shown on a logarithmic scale.

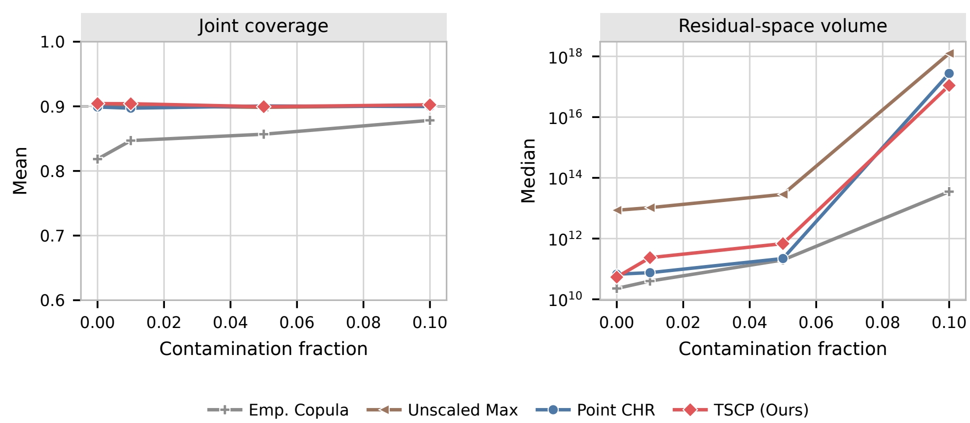


Figure S6: Row-wise contamination stress test with $d = 10$ and $n = 100$. A contaminated observation has every noise coordinate multiplied by 10. The right panel reports median residual-space volume over 200 repetitions on a logarithmic scale.

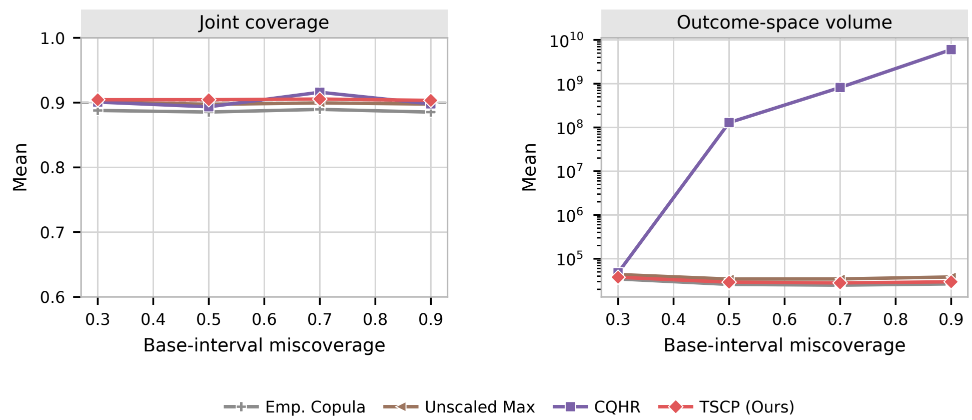


Figure S7: CQR sensitivity to the base-interval miscoverage with $d = 3$, $n = 100$, and final target coverage 0.9. TSCP uses capped scores, CQHR uses its native construction, and results average 30 independent repetitions.

C.7 Additional CQR Sensitivity Analyses

Base-interval width. The main CQR comparison uses base-interval miscoverage 0.5. We repeat it at 0.3, 0.5, 0.7, and 0.9, fixing $n = 100$ and final miscoverage 0.1. Figure S7 shows that capped-score TSCP has empirical coverage between 0.903 and 0.906 and mean outcome-space volume between 2.80×10^4 and 3.77×10^4 across the sweep. Unscaled Max behaves similarly, while Emp. Copula undercovers. CQHR remains close to nominal coverage but its volume is highly sensitive to the fitted base interval in this design. The result supports using the moderate 0.5 base interval for the primary comparison and cautions against drawing conclusions from one extreme base-quantile choice.

Shifted scores. An alternative way to enforce nonnegativity is to replace the signed CQR score S_j by $S_j + C$, apply TSCP, and subtract C from the final coordinate adjustment. We test $C \in \{100, 300, 1000\}$. As shown in Figure S8, the recorded TSCP rectangle is invariant to C in this range up to numerical roundoff and agrees exactly, to recorded numerical precision, with the TSCP-GWC rectangle: the maximum coordinate-length difference is zero in all 30 repetitions for every C. Consequently, this experiment does not identify a practically meaningful shift constant and the shifted construction does not provide a distinct local refinement in this design. This is an empirical finding, not a translation-invariance theorem for arbitrary data. A fixed shift is justified by a bound on the score over the relevant input domain, not only its observed calibration minimum: for ordered base intervals of width $w_j(x)$, $S_j(x, y) \geq -w_j(x)/2$, so a known uniform width bound $w_j(x) \leq M$ gives the sufficient choice $C \geq M/2$. The experiment uses fixed constants and is not a general selector for C. We therefore retain the capped nonnegative score in the main text and do not recommend tuning C.

C.8 Low-Dimensional Shape-Template Baseline

Finally, we compare with the convex shape-template approach of Tumu et al. (2024). We use the public `conformal-region-designer` components: a kernel density estimate on a grid of

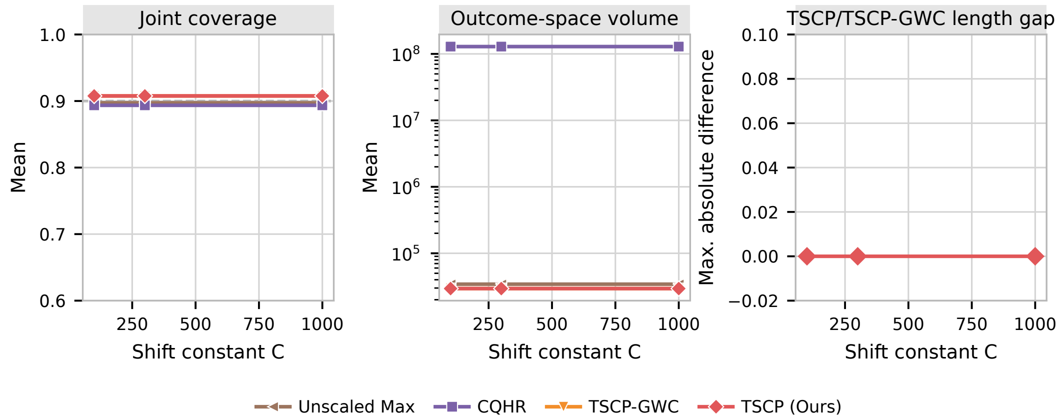


Figure S8: Sensitivity of shifted CQR scores to the additive constant C. The first two panels report coverage and mean outcome-space volume; the third reports the maximum coordinate-length difference between shifted TSCP and TSCP-GWC. Results use $d = 3$, $n = 100$, and 30 independent repetitions.

size 20 per dimension, mean-shift clustering, and an axis-aligned hyperrectangular template. The signed calibration residuals are divided equally between fitting the density and template and conformalizing its inflation. We use a score-consistent hyperrectangle implementation and impose an aspect-ratio floor of 0.05 when a very small fitted cluster is degenerate along one coordinate. This detail keeps the conformal inflation finite and is stated here because it differs from calling the released rectangle class without modification.

We use $d = 2$ with independent Gaussian scales $(2, 1)$ and calibration sizes $n \in \{30, 50, 100, 200\}$. Shape-template volume is divided by 2^d to match the residual-space volume convention used for the symmetric rectangular methods. Figure S9 shows broadly comparable coverage. At $n = 100$, the shape-template method has coverage 0.904, mean normalized volume 10.07, and mean construction time 98.7 milliseconds; TSCP has coverage 0.906, mean volume 8.04, and mean construction time 0.80 milliseconds. For each sample size shown, TSCP has a smaller mean normalized volume. Its construction time is also over two orders of magnitude shorter for this implementation.

This is deliberately a low-dimensional comparison. A Cartesian KDE grid has 20^d locations and is therefore not computationally meaningful at the $d = 10$ scale of the main synthetic experiment. The experiment establishes a comparison with a flexible convex template where the released computational strategy is applicable; it is not evidence that either method dominates all shape-template choices in higher dimensions.

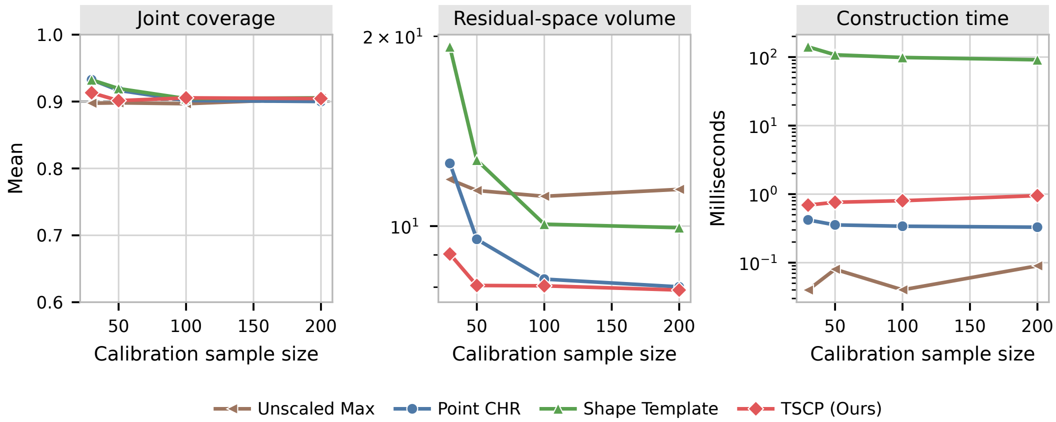


Figure S9: Comparison with a KDE/mean-shift hyperrectangular shape template for $d = 2$ Gaussian residuals with scales $(2, 1)$. Volume follows the same 2^{-d} normalization as the other residual-space experiments. Results average 100 independent repetitions, and runtime excludes the common regression fit.

C.9 Real-Data Coordinate Coverage and Uncertainty

The new real-data diagnostic runs in Section 4.5 retain the original split proportions, calibration sizes, and model settings. In each of the 200 splits we record full interval lengths and marginal test coverage for all 51 output coordinates across the six datasets, for all four methods. Together this gives 40,800 split–method–coordinate records. These are paired comparisons on common fitted models, not separate favorable splits selected for each method.

Figure S10 reports all coordinate-wise coverage means, and Table S4 reports the matching joint coverage. The 0.9 line is the joint target used for calibration, not an assertion that every marginal should equal 0.9. For a rectangular region, each coordinate's empirical coverage is at least the empirical joint coverage on the same test set, but the marginal rates can differ substantially. Thus, shorter intervals in Figure 9 must be interpreted alongside both kinds of coverage.

TSCP's mean joint coverages are 0.956, 0.902, 0.901, 0.901, 0.923, and 0.912 on `stock`, `rf2`, `scm1d`, `scm20d`, `energy`, and `student`, respectively. The target lies within one across-split standard deviation of each mean; this is a variability diagnostic, not a confidence interval for the mean. The coordinate means are not equal: for example, TSCP's marginal coverages range from about 0.972 to 0.999 on `stock`, while its joint coverage is 0.956. On `energy`, the two marginal coverages are 0.962 and 0.954, with joint coverage 0.923. This illustrates why interval-length allocation and marginal balance cannot be read off from a single joint-coverage number.

The accompanying data directory `experiment_data/real_diagnostics` includes raw lengths, paired length ratios, marginal and joint coverages, standard deviations, target names, and the exact split identifiers. Standard errors obtained by dividing the across-split standard deviation by $\sqrt{200}$ quantify Monte Carlo variation over randomized splits of each fixed

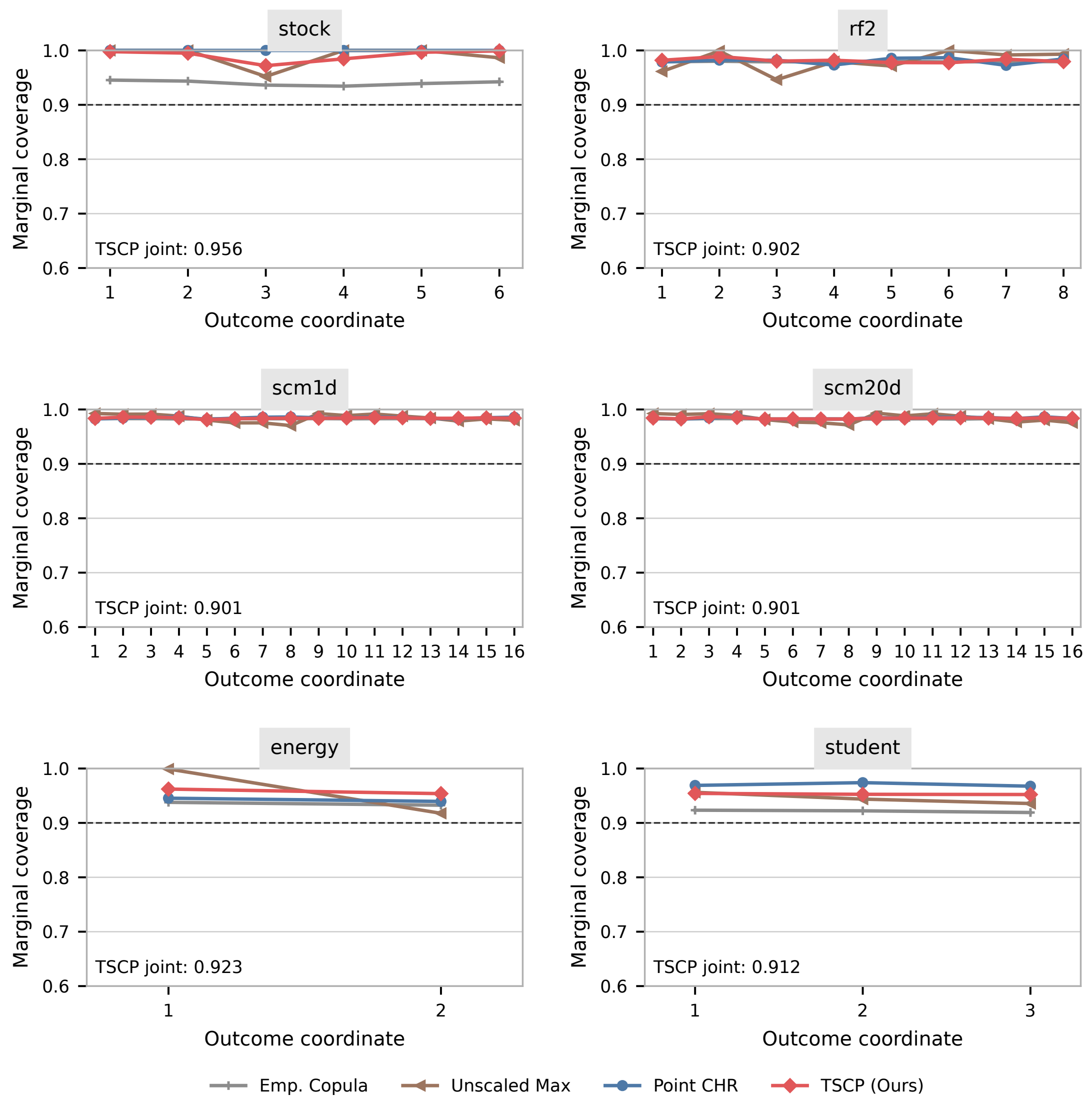


Figure S10: Coordinate-wise marginal coverage over 200 new splits of each real dataset. The dashed line is the 0.9 joint target, and each panel also reports TSCP's mean joint coverage. Axes retain the range $[0.6, 1.0]$. Across-split standard deviations and pointwise Monte Carlo intervals are retained in the supporting CSV files rather than drawn as error bars. Coordinate order is the original target-column order.

Dataset	Emp. Copula	Unscaled Max	Point CHR	TSCP
`stock`	0.727 (0.113)	0.940 (0.057)	1.000 (0.000)	0.956 (0.059)
`rf2`	0.893 (0.016)	0.900 (0.016)	0.901 (0.022)	0.902 (0.017)
`scm1d`	0.893 (0.016)	0.900 (0.016)	0.905 (0.020)	0.901 (0.015)
`scm20d`	0.892 (0.017)	0.901 (0.016)	0.902 (0.023)	0.901 (0.016)
`energy`	0.887 (0.061)	0.917 (0.048)	0.899 (0.069)	0.923 (0.048)
`student`	0.861 (0.073)	0.904 (0.055)	0.939 (0.057)	0.912 (0.056)

Table S4: Joint test coverage in the new real-data diagnostic cohort. Entries are the mean and one across-split standard deviation over 200 reruns. These measurements accompany the new coordinate plots and do not replace the preserved historical real-data table.

dataset. They do not measure uncertainty over independently sampled populations, and the reported pointwise intervals are not simultaneous statements over all coordinates.

C.10 Search-Regime Stress Tests on Cached Real Residuals

To explore regimes beyond the main 90% target, we reuse the same 200 calibration residual matrices per dataset and vary the miscoverage level over $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. This is a computational diagnostic, not an additional 90%-coverage experiment: the nominal coverage changes with α. No model is refitted or selected based on the branch outcomes. The sweep records 6,000 dataset–split–level configurations. It separates runs entering any backward search from those taking the global fallback.

Unlike the main target, the sweep reaches both exceptional branches. On `energy` at $\alpha = 0.5$, backward search occurs in 3/200 splits (1.5%), while fallback occurs in 140/200 (70%). The remaining 57 splits use only binary search. Mean whole-call times conditional on these three regimes are 0.295, 0.121, and 0.268 ms, respectively. At $\alpha = 0.7$ and 0.9, 10 energy splits use backward search and 190 fall back. In every observed backward-search case, each of the two coordinates inspects only one backward candidate. These examples therefore exercise the branch without exhibiting a long backward scan.

At $\alpha = 0.9$, fallback counts are 139, 185, 180, 186, 190, and 200 out of 200 for `stock`, `rf2`, `scm1d`, `scm20d`, `energy`, and `student`, respectively. No backward search is observed in the other five datasets over the displayed levels. Fallback is often faster because it returns the already computed global rectangle, so a lower average runtime at an extreme target need not indicate a more efficient local search. The saved summaries include counts, coordinate denominators, candidate-cell counts, and conditional times for every dataset–level pair; an unobserved conditional time remains missing.

Search-branch indicators are saved separately from returned rectangles: equality between TSCP and TSCP-GWC output is not treated as proof of a fallback. Optional instrumentation was checked against the pre-instrumentation implementation, including explicit binary, backward, and fallback cases. All 1,200 default-target real-data rectangles also match that implementation exactly. This study measures the executed implementation and does not resolve the separate proof and endpoint-inequality questions in the methodology.

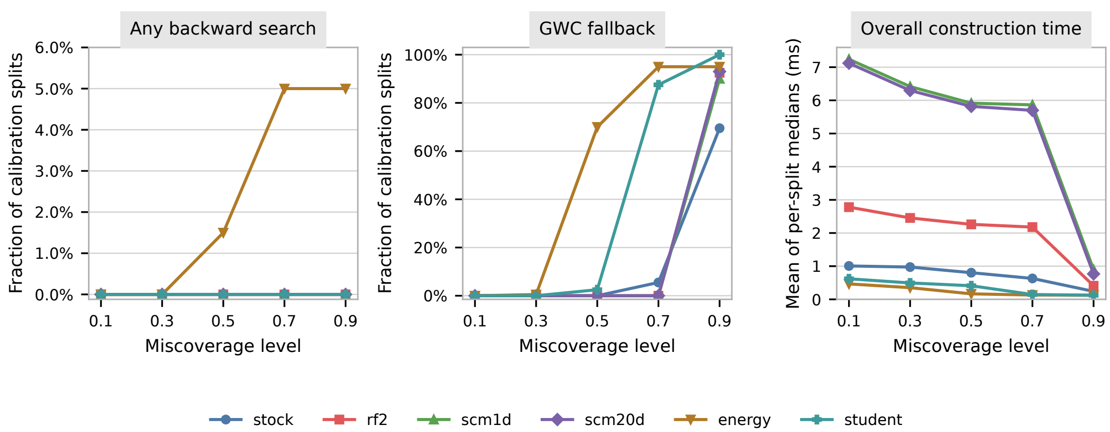


Figure S11: Computational stress sweep on the cached real-data calibration residuals. Each point summarizes 200 splits. The first two panels show the fractions of splits using any backward search or the global fallback; their vertical ranges differ. The final panel reports construction time. Additional miscoverage levels use the median of three uninstrumented calls per split; the main $\alpha = 0.1$ results use seven. All timing summaries average these per-split medians.

Appendix D. Retained Original Numerical Experiments

This appendix retains every original experimental figure and table from the supplied manuscript, apart from the two-point energy example, which remains in the main text as Table 2. The results below are historical comparisons, not additional fresh-draw repetitions of the new experiments. They complement Appendix C and are kept to preserve the original evidence, including comparisons favorable to competing methods.

D.1 Methods and Metrics

We provide a comprehensive empirical validation of the proposed *TSCP* method (Algorithm 2) using simulated and real data. The performance of TSCP is compared to that of three representative benchmark approaches: (i) *Unscaled Max*, a direct extension of split conformal inference that applies the ℓ_∞ norm to the residuals, summarized in Algorithm S2; (ii) *Point CHR*, a data-splitting variant recently proposed by Sampson and Chan (2024), which estimates per-target out-of-sample points using a held-out calibration set and adjusts them to form axis-aligned rectangular sets; (iii) *Emp. Copula*, the first distribution-free copula-based method introduced in Messoudi et al. (2021).

The results of additional experiments comparing TSCP to different approaches are in Appendix C. We do not include in this empirical study alternative copula-based methods—e.g., vine copulas (Park and Cho, 2025)—because their main limitations relative to our approach are the same as those of the *Emp. Copula* benchmark. To enable meaningful direct comparisons, we also omit approaches that do not produce rectangular prediction sets.

For simplicity, we apply all methods using the same conditional mean-regression model $\hat{f}$ and standard absolute residuals; i.e., $E_j(\mathbf{X}, Y_j) = |Y_j - \hat{f}_j(\mathbf{X})|$ for all $j \in [d]$. Although our method can also be seamlessly applied using different choices of models and residuals, this is a relatively common choice previously employed in several related works (Sampson and Chan, 2024; Park and Cho, 2025; Dheur et al., 2025).

The conformal prediction sets produced by the different methods are compared based on two key performance metrics: average coverage and size (volume). Since all methods rely on the same underlying model, it suffices to measure the size of the prediction sets in residual space. Specifically, we report

$$\text{(Test) Coverage} := \frac{1}{N}\sum_{l=1}^{N} \frac{\#\text{ of included test points}}{\#\text{ of total test points}},$$

$$\text{(Residual space) Volume} := \frac{1}{N}\sum_{l=1}^{N} \frac{1}{2^d}\text{Vol(prediction set)},$$

where both metrics are averaged over N repetitions of each experiment, under the original resampling protocol. The constant factor 2^{-d} in the volume definition facilitates comparison across experiments with different output dimensions.

D.2 Experiments with Simulated Data

We simulate data with features from a 10-dimensional standard normal distribution with independent components, and 10-dimensional outcomes

$$Y_j \mid X \sim f_j(X) + \epsilon_j, \quad \forall j \in [10], \tag{S4}$$

where $f_j(X) = \sum_{i=1}^{10} \xi_i X_i$ such that $\xi_1, \ldots, \xi_{10} \sim \text{Unif}(-10, 10)$, and ϵ_j is independent random noise for each $j \in [d]$. We consider six settings corresponding to different noise distributions; the results for two are presented here while those corresponding to other settings, which lead to qualitatively similar conclusions, are summarized in Appendix D.4.1.

The original generating implementation forms a pool of 8,000 observations and uses randomized 80%/20% training/test splits, giving 6,400 training and 1,600 test observations per split. A separate calibration sample has size $|\mathcal{D}_{\text{cal}}| \in \{30, 50, 100, 300, 500\}$, and $\alpha = 0.1$. The results average $N = 200$ repetitions of that original protocol. Because the training/test pool is reused, these should not be interpreted as 200 independent population draws. In contrast, the revised experiments in Section 4 redraw 7,200 training and 800 test observations in every repetition, matching the allocation described in the original manuscript. The historical figures are retained unchanged; only their protocol description is corrected here.

Figure S12 summarizes the results obtained in the first setting, where the noise distribution is Gaussian with heterogeneous variance across dimensions:

$$\epsilon_j^{\text{heter}} \sim \mathcal{N}\left(0, \ (10 - j + 1)^2\right), \quad \forall j \in [10].$$

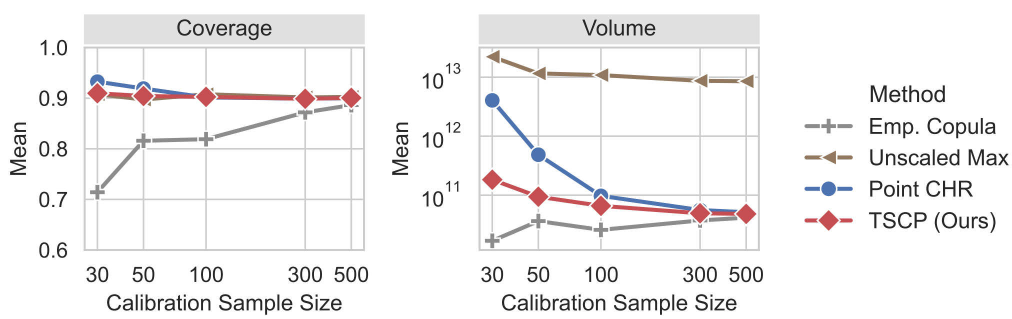


Figure S12: Average performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables and heterogeneous noise variances. The target coverage level is 90%. TSCP consistently attains near-nominal coverage while producing the smallest prediction sets on average. Numerical results and corresponding standard deviations are reported in Table S6.

These results show that our method performs well across both small and large sample sizes, with empirical coverage close to $1 - \alpha$, using relatively tight prediction sets. The *Point CHR* approach yields substantially larger, and therefore less informative, prediction sets—especially in small samples (e.g., $n < 100$)—due to the inefficiency introduced by its additional sample split. Even for moderately large samples, our method continues to perform noticeably better, although this difference is difficult to discern in Figure S12 because of the log–log scale; see Table S6 for a more detailed numerical summary. The *Emp. Copula* approach clearly fails to achieve finite-sample coverage, since it uses the same calibration set both to estimate the copula model and to calibrate the prediction sets based on the resulting nonconformity scores. While this issue could be addressed through additional data splitting, doing so would come with the same efficiency disadvantages of *Point CHR*. This limitation is shared by related copula-based approaches that we do not explicitly include in our comparisons. Finally, the *Unscaled Max* approach produces much larger (and effectively uninformative) prediction sets in both small and large samples, as it is negatively affected by the unequal variances of the noise distribution across different dimensions.

Figure S13 reports results from analogous experiments in which the noise distribution has equal variance across dimensions, with $\epsilon_j^{\text{homo}} \sim \mathcal{N}(0,1)$ for all $j \in [10]$. This setting strongly favors the *Unscaled Max* approach; nevertheless, our TSCP still performs as a close second. In practice, however, noise levels are rarely known *a priori*, making the adaptivity and robustness of TSCP generally desirable.

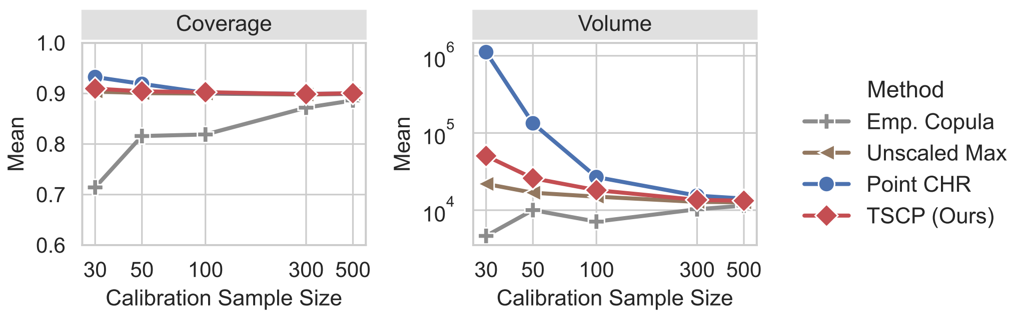


Figure S13: Average performance of the proposed TSCP method and alternative conformal prediction approaches in numerical experiments similar to those of Figure S12. Here, the outcome variables have homogeneous variance across their $d = 10$ dimensions, which gives an advantage to the *Unscaled Max* approach. The target coverage level is 0.9. TSCP attains near-nominal coverage and is competitive, while Unscaled Max is the most efficient in this homogeneous setting. Numerical results and corresponding standard deviations are reported in Table S7.

The results of additional numerical experiments are reported in Appendices D.5–D.6. In Appendix D.5, we compare TSCP with several alternative approaches described in Sections 2 and 3, including the naive plug-in rescaling procedure (Algorithm S3), the data-splitting variant (Algorithm S4), the conservative global worst-case approach (Algorithm S5), and the union-based construction without rectangular enclosure (Section 3.4). These experiments highlight the favorable balance between statistical–computational trade-offs achieved by TSCP. We further examine scalability with respect to the outcome dimension and show that Algorithm 2 remains computationally efficient in regimes where union-based constructions become prohibitive. In Appendix D.6, we investigate the robustness of TSCP under heavy-tailed noise and demonstrate that our method has near-nominal empirical coverage in the displayed infinite-variance cases; this is a stress test outside Assumption 1, not a theoretical extension.

D.3 Experiments with Real Data

We now evaluate all methods on six real datasets. The first four, *rf2* (Tsoumakas et al., 2022), *scm1d* (Contributors, 2019a), *scm20d* (Contributors, 2019b), and *stock* (Yeh, 2015), are borrowed from related prior works (Park and Cho, 2025; Dheur et al., 2025). The other two datasets, *student* (Cortez, 2008) and *energy* (Tsanas and Xifara, 2012), are obtained from the UCI repository and serve to illustrate the small-sample performance of TSCP.

For each dataset, we perform 200 random train/calibration/test splits using a 75%/5%/20% ratio. We fit a multivariate sparse (lasso) regression model with regularization parameter $\lambda = 0.0001$ on *stock*, and random forest regression models on all other datasets; see the GitHub software repository linked in the Software Availability section for additional reproducibility details. Table S5 reports the average coverage and volume, along with one standard deviation. As shown in Table S5, the empirical behavior of all methods on real data largely mirrors the trends observed in the simulation studies. Among methods with theoretical coverage guarantees, TSCP achieves the smallest prediction-set volume on the *stock*, *scm1d*, *scm20d*, and *energy* datasets. On *rf2*, TSCP is outperformed by *Point CHR*, likely due to the presence of a small number of influential outliers. On *student*, *Unscaled Max* yields smaller volumes, consistent with the approximately homogeneous noise across targets (student GPAs at different grade levels). The *Point CHR* approach produces an uninformative, *infinite* prediction set on the *stock* dataset, because its additional data split makes the effective calibration set extremely small. Finally, *Emp. Copula* generally fails to achieve the desired coverage level, consistent with its lack of finite-sample guarantees. Overall, the results show that TSCP provides a reasonable trade-off between coverage and volume, excelling especially when calibration data are limited or when noise scales differ significantly across targets.

Method	Stock ($d = 6$, $n = 15$)		rf2 ($d = 8$, $n = 384$)	
	Coverage	Volume	Coverage	Volume
Unscaled Max	0.940(0.057)	6.03e-02(1.57e-01)	0.900(0.016)	4.14e+03(2.90e+03)
Emp. copula	0.727(0.113)	1.01e-05(9.80e-06)	0.893(0.016)	5.35e+01(4.69e+01)
Point CHR	1.000(0.000)	inf	0.901(0.022)	**6.85e+01(7.55e+01)**
TSCP	0.956(0.059)	**1.86e-03(8.55e-03)**	0.901(0.017)	5.42e+02(1.50e+03)

Method	scm1d ($d = 16$, $n = 490$)		scm20d ($d = 16$, $n = 448$)	
	Coverage	Volume	Coverage	Volume
Unscaled Max	0.900(0.016)	2.74e+39(4.02e+39)	0.901(0.016)	2.70e+40(2.92e+40)
Emp. copula	0.893(0.016)	1.09e+39(1.34e+39)	0.892(0.017)	1.178e+40(1.190e+40)
Point CHR	0.905(0.020)	2.84e+39(5.43e+39)	0.902(0.023)	3.00e+40(8.00e+40)
TSCP	0.901(0.015)	**1.52e+39(2.61e+39)**	0.901(0.016)	**1.53e+40(1.50e+40)**

Method	Energy ($d = 2$, $n = 38$)		Student ($d = 3$, $n = 32$)	
	Coverage	Volume	Coverage	Volume
Unscaled Max	0.917(0.049)	1.58e+01(6.80e+00)	0.904(0.055)	**1.51e+02(1.27e+02)**
Emp. copula	0.886(0.061)	5.41e+00(4.49e+00)	0.861(0.073)	1.08e+02(7.19e+01)
Point CHR	0.899(0.069)	8.81e+00(2.42e+01)	0.939(0.057)	6.89e+02(1.07e+03)
TSCP	0.922(0.048)	**6.95e+00(4.43e+00)**	0.912(0.056)	1.98e+02(1.77e+02)

Table S5: Performance of the proposed TSCP method and alternative conformal prediction approaches on real datasets. Results are averaged over 200 random splits of the data into training/calibration/and test subsets, with one standard deviation reported in parentheses. The target coverage level is 0.9. Coverage values below the target are highlighted in red, and the smallest volume among methods achieving valid coverage is highlighted in **bold**.

D.4 Retained Distribution-Specific Results

This section presents the results of additional numerical experiments, including comparisons with a broader range of alternative approaches. These experiments are conducted using the same protocol and performance metrics described in Appendix D unless specified otherwise.

D.4.1 Distribution-Specific Tables

Here we present additional results from the numerical experiments described in Appendix D, corresponding to five noise settings. These results are qualitatively consistent with those summarized in Appendix D: our method performs consistently well, producing relatively tight prediction sets with near-nominal empirical coverage.

Gaussian Noise (heterogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gaussian)	Emp. Copula	0.714 (0.079)	1.68e+10(1.16e+10)
	Point CHR	0.933 (0.066)	4.04e+12(1.38e+13)
	TSCP (Ours)	0.910 (0.053)	**1.83e+11(2.74e+11)**
	Unscaled Max	0.908 (0.047)	2.22e+13(4.77e+13)
(50, 10, Gaussian)	Emp. Copula	0.816 (0.057)	3.64e+10(2.22e+10)
	Point CHR	0.919 (0.057)	4.85e+11(1.72e+12)
	TSCP (Ours)	0.904 (0.047)	**9.42e+10(8.82e+10)**
	Unscaled Max	0.897 (0.043)	1.16e+13(1.10e+13)
(100, 10, Gaussian)	Emp. Copula	0.819 (0.039)	2.58e+10(9.90e+09)
	Point CHR	0.901 (0.041)	9.74e+10(9.22e+10)
	TSCP (Ours)	0.903 (0.034)	**6.59e+10(3.43e+10)**
	Unscaled Max	0.908 (0.027)	1.09e+13(6.93e+12)
(300, 10, Gaussian)	Emp. Copula	0.872 (0.019)	3.73e+10(8.11e+09)
	Point CHR	0.899 (0.025)	5.60e+10(2.05e+10)
	TSCP (Ours)	0.899 (0.018)	**4.93e+10(1.17e+10)**
	Unscaled Max	0.901 (0.020)	8.68e+12(3.29e+12)
(500, 10, Gaussian)	Emp. Copula	0.886 (0.018)	4.16e+10(8.27e+09)
	Point CHR	0.900 (0.022)	5.13e+10(1.49e+10)
	TSCP (Ours)	0.901 (0.016)	**4.81e+10(9.67e+09)**
	Unscaled Max	0.903 (0.016)	8.54e+12(2.46e+12)

Table S6: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, as in Figure S12. All results are averaged over 200 trials with their one standard deviation reported in parenthesis. The target coverage level is 0.9. Coverage below this target is highlighted in red. For volume, lower is better. Smallest volume size (among those with valid theoretical coverage) is highlighted in **bold**. TSCP yields the tightest volume while maintaining valid coverage.

Gaussian Noise (homogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Unit Gaussian)	Emp. Copula	0.714 (0.079)	4.64e+03(3.20e+03)
	Point CHR	0.933 (0.066)	1.11e+06(3.80e+06)
	TSCP (Ours)	0.910 (0.053)	5.04e+04(7.56e+04)
	Unscaled Max	0.904 (0.054)	**2.19e+04(1.94e+04)**
(50, 10, Unit Gaussian)	Emp. Copula	0.816 (0.057)	1.00e+04(6.12e+03)
	Point CHR	0.919 (0.057)	1.34e+05(4.75e+05)
	TSCP (Ours)	0.904 (0.047)	2.60e+04(2.43e+04)
	Unscaled Max	0.901 (0.045)	**1.68e+04(1.15e+04)**
(100, 10, Unit Gaussian)	Emp. Copula	0.819 (0.039)	7.11e+03(2.73e+03)
	Point CHR	0.901 (0.041)	2.68e+04(2.54e+04)
	TSCP (Ours)	0.903 (0.034)	1.82e+04(9.46e+03)
	Unscaled Max	0.900 (0.035)	**1.50e+04(7.38e+03)**
(300, 10, Unit Gaussian)	Emp. Copula	0.872 (0.019)	1.03e+04(2.23e+03)
	Point CHR	0.899 (0.025)	1.54e+04(5.66e+03)
	TSCP (Ours)	0.899 (0.018)	1.36e+04(3.23e+03)
	Unscaled Max	0.898 (0.018)	**1.28e+04(3.07e+03)**
(500, 10, Unit Gaussian)	Emp. Copula	0.886 (0.018)	1.15e+04(2.28e+03)
	Point CHR	0.900 (0.022)	1.41e+04(4.11e+03)
	TSCP (Ours)	0.901 (0.016)	1.33e+04(2.67e+03)
	Unscaled Max	0.899 (0.017)	**1.26e+04(2.50e+03)**

Table S7: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, as in Figure S13. Other details are as in Table S6. In this case, Unscaled Max yields the tightest volume while maintaining valid coverage, while TSCP remains a competitive second.

Laplace Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Laplace)	Emp. Copula	0.712 (0.079)	3.82e+12(4.91e+12)
	Point CHR	0.938 (0.060)	5.57e+16(2.71e+17)
	TSCP (Ours)	0.908 (0.060)	**3.14e+14(9.93e+14)**
	Unscaled Max	0.897 (0.053)	6.36e+15(1.99e+16)
(50, 10, Laplace)	Emp. Copula	0.819 (0.056)	1.36e+13(1.75e+13)
	Point CHR	0.917 (0.050)	6.64e+14(1.81e+15)
	TSCP (Ours)	0.906 (0.045)	**7.32e+13(1.60e+14)**
	Unscaled Max	0.901 (0.041)	3.34e+15(5.12e+15)
(100, 10, Laplace)	Emp. Copula	0.811 (0.039)	5.96e+12(3.94e+12)
	Point CHR	0.896 (0.039)	4.74e+13(6.07e+13)
	TSCP (Ours)	0.898 (0.030)	**2.64e+13(1.94e+13)**
	Unscaled Max	0.895 (0.029)	1.77e+15(1.78e+15)
(300, 10, Laplace)	Emp. Copula	0.871 (0.020)	1.11e+13(4.03e+12)
	Point CHR	0.895 (0.023)	2.01e+13(1.12e+13)
	TSCP (Ours)	0.896 (0.017)	**1.74e+13(6.40e+12)**
	Unscaled Max	0.896 (0.017)	1.38e+15(7.08e+14)
(500, 10, Laplace)	Emp. Copula	0.882 (0.016)	1.29e+13(4.10e+12)
	Point CHR	0.896 (0.021)	1.83e+13(8.95e+12)
	TSCP (Ours)	0.896 (0.016)	**1.64e+13(5.68e+12)**
	Unscaled Max	0.897 (0.017)	1.36e+15(5.51e+14)

Table S8: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Laplace* distribution; i.e., $\epsilon_j \sim \text{Laplace}(10 - j + 1)$ for all $j \in [10]$. Other details are as in Table S6.

Mixture Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Mixed)	Emp. Copula	0.715 (0.075)	2.69e+11(3.04e+11)
	Point CHR	0.935 (0.062)	2.56e+15(1.98e+16)
	TSCP (Ours)	0.913 (0.051)	**9.53e+12(3.12e+13)**
	Unscaled Max	0.903 (0.052)	3.46e+13(6.74e+13)
(50, 10, Mixed)	Emp. Copula	0.821 (0.058)	7.07e+11(5.96e+11)
	Point CHR	0.925 (0.048)	6.10e+13(3.06e+14)
	TSCP (Ours)	0.906 (0.044)	**2.57e+12(3.91e+12)**
	Unscaled Max	0.901 (0.044)	1.98e+13(1.68e+13)
(100, 10, Mixed)	Emp. Copula	0.819 (0.035)	3.82e+11(1.98e+11)
	Point CHR	0.898 (0.042)	2.76e+12(4.47e+12)
	TSCP (Ours)	0.900 (0.029)	**1.23e+12(7.54e+11)**
	Unscaled Max	0.905 (0.028)	1.65e+13(9.89e+12)
(300, 10, Mixed)	Emp. Copula	0.874 (0.022)	6.47e+11(2.25e+11)
	Point CHR	0.902 (0.025)	1.27e+12(7.23e+11)
	TSCP (Ours)	0.900 (0.019)	**9.05e+11(3.48e+11)**
	Unscaled Max	0.900 (0.020)	1.36e+13(5.22e+12)
(500, 10, Mixed)	Emp. Copula	0.886 (0.016)	7.26e+11(1.82e+11)
	Point CHR	0.897 (0.020)	1.01e+12(5.42e+11)
	TSCP (Ours)	0.898 (0.014)	**8.17e+11(2.05e+11)**
	Unscaled Max	0.902 (0.016)	1.35e+13(3.73e+12)

Table S9: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Mixture* distribution; i.e., $\epsilon_j \sim \frac{1}{2}\text{Laplace}(10 - j + 1) + \frac{1}{2}\mathcal{N}(0, (10 - j + 1)^2)$ for all $j \in [10]$. Other details are as in Table S6.

Gamma Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gamma)	Emp. Copula	0.714 (0.067)	5.33e-02(6.85e-02)
	Point CHR	0.929 (0.058)	7.55e+02(4.32e+03)
	TSCP (Ours)	0.904 (0.056)	**3.77e+00(1.59e+01)**
	Unscaled Max	0.898 (0.051)	2.69e+15(6.18e+15)
(50, 10, Gamma)	Emp. Copula	0.826 (0.048)	2.20e-01(2.78e-01)
	Point CHR	0.927 (0.049)	5.01e+01(3.36e+02)
	TSCP (Ours)	0.910 (0.041)	**1.15e+00(2.56e+00)**
	Unscaled Max	0.902 (0.039)	1.94e+15(3.54e+15)
(100, 10, Gamma)	Emp. Copula	0.818 (0.038)	1.04e-01(1.01e-01)
	Point CHR	0.906 (0.038)	1.28e+00(2.38e+00)
	TSCP (Ours)	0.900 (0.030)	**4.65e-01(5.82e-01)**
	Unscaled Max	0.897 (0.030)	9.67e+14(7.86e+14)
(300, 10, Gamma)	Emp. Copula	0.873 (0.019)	1.75e-01(1.15e-01)
	Point CHR	0.895 (0.024)	3.74e-01(3.70e-01)
	TSCP (Ours)	0.898 (0.018)	**2.67e-01(1.69e-01)**
	Unscaled Max	0.895 (0.018)	7.04e+14(3.45e+14)
(500, 10, Gamma)	Emp. Copula	0.885 (0.015)	1.89e-01(1.06e-01)
	Point CHR	0.896 (0.020)	3.20e-01(2.33e-01)
	TSCP (Ours)	0.897 (0.016)	**2.36e-01(1.35e-01)**
	Unscaled Max	0.893 (0.015)	6.37e+14(2.53e+14)

Table S10: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Gamma* distribution; i.e., $\epsilon_j \sim \text{Gamma}(1, 10 - j + 1)$ for all $j \in [10]$. Other details are as in Table S6.

D.5 Comparison of Alternative Implementations of TSCP

The additional methods considered in this section include (i) *Naive*, the heuristic plug-in rescaling method that uses location and scale parameters estimated directly from the calibration data without further adjustments, summarized in Algorithm S3; (ii) *TSCP-S*, the inefficient data-splitting variant summarized in Algorithm S4; (iii) *TSCP-GWC*, our conservative global-worst-case method summarized in Algorithm S5; (iv) *TSCP-LWC*, the aggregated union described in Section 3.4 but not enclosed by a rectangle.

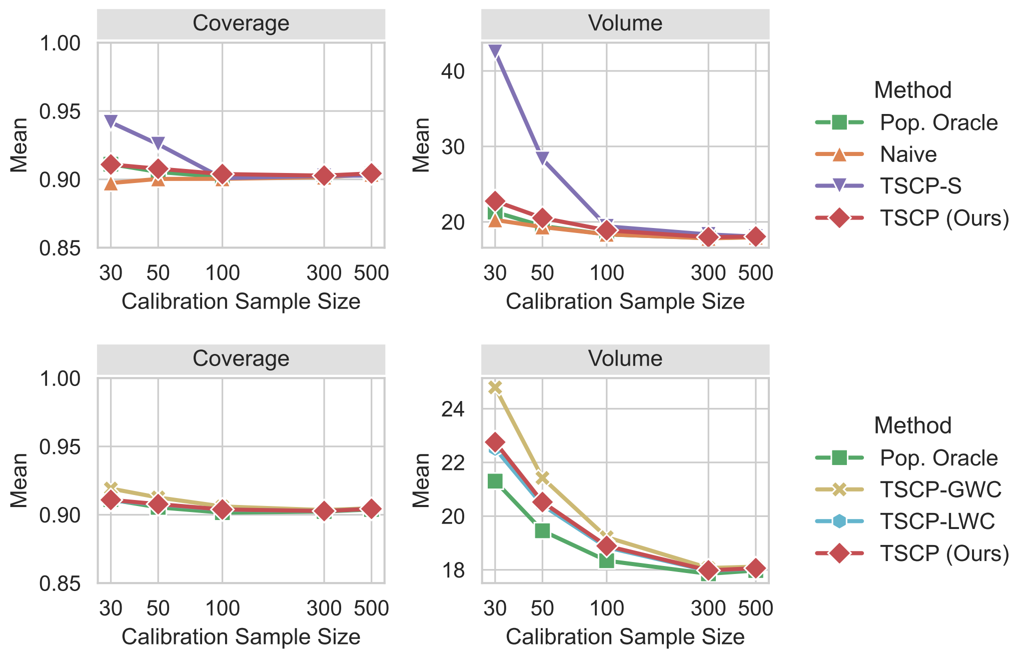


Figure S14: Performance of TSCP and alternative approaches on simulated data with $d = 2$ targets and heterogeneous Laplace noise. The target coverage level is 90%. Results are averaged over 200 trials. All methods achieve near-nominal coverage and approximate *Pop. Oracle* as the calibration sample size increases. TSCP tracks TSCP-LWC closely, with an almost indistinguishable average volume, highlighting the tightness of its rectangular enclosure.

All methods compared in Figure S14 tend to approximate *Pop. Oracle* well as n increases, with TSCP and TSCP-LWC being the two most stable and fastest approximations among all methods with theoretical coverage. These results also show that TSCP encloses the aggregated union TSCP-LWC very tightly, and that TSCP-GWC is conservative, though it still performs better than the data-splitting variant TSCP-S. Despite lacking a finite-sample coverage guarantee, the *Naive* approach performs well, as also shown in Figure S15.

Figure S16 better highlights the prohibitive computational cost of TSCP-LWC, by reporting on the results of similar experiments where a small sample size is fixed, $|\mathcal{D}_{\text{cal}}| = 30$, while the outcome dimensions are gradually increased: $d \in \{2, 3, 4, 5, 10, 20, 30\}$.

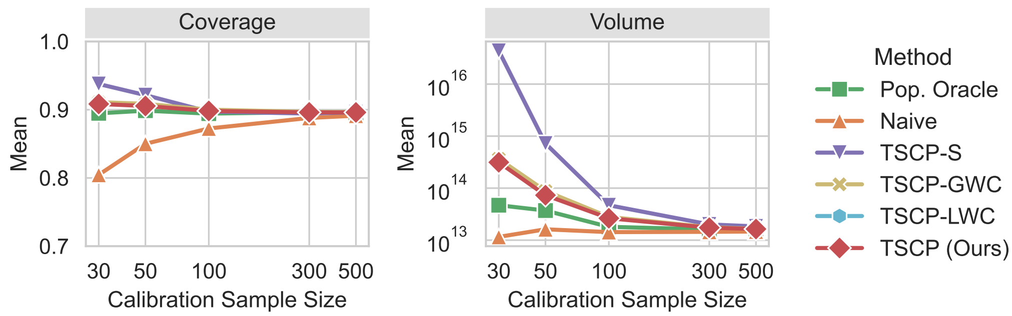


Figure S15: Performance of TSCP and alternative approaches on simulated data with $d = 10$ targets. Other details are as in Figure S14. TSCP-LWC is omitted from this figure due to its prohibitive cost with even moderately high-dimensional outcomes.

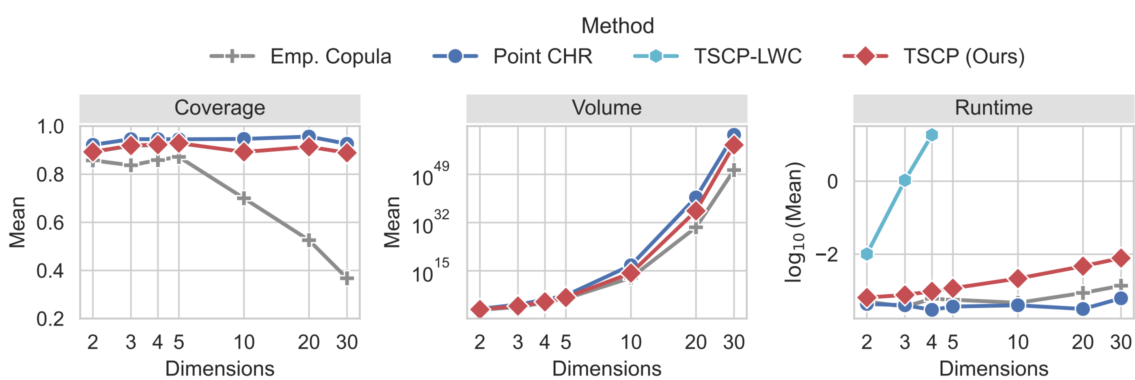
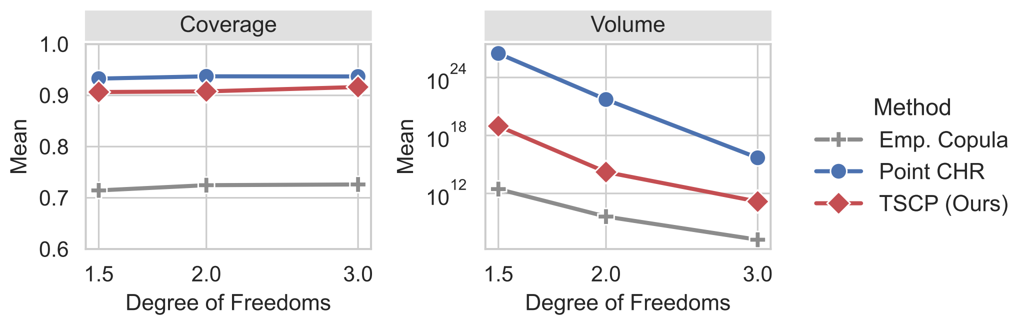


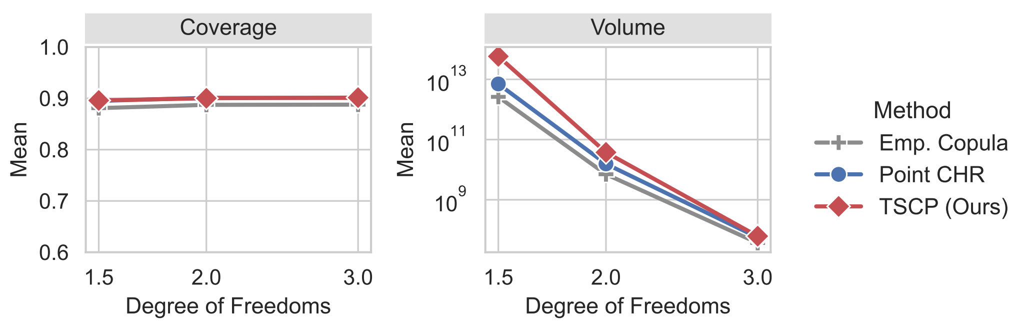
Figure S16: Performance of TSCP and alternative approaches on simulated data with fixed sample size $|\mathcal{D}_{\mathrm{cal}}| = 30$, as a function of the number of output dimensions d. Other details are as in Figure S14. These results are averaged over 30 repetitions under the original resampling protocol. The runtime results are presented in seconds on a $\log_{10}$ scale. The runtime of all methods except TSCP-LWC scales well with number of dimensions. TSCP-LWC is only applied when $d \in \{2, 3, 4\}$ due to the exponential growth of its runtime.

D.6 Robustness to Heavy-Tailed Data

We test the performance of TSCP on simulated heavy-tailed data with $d = 10$ outcomes and noise following a Student's-t distribution; i.e., $Y_j \mid \mathbf{X} \sim f_j(\mathbf{X}) + \epsilon_j$ with $\epsilon_j \sim t_{\mathrm{DF}}$ and $\mathrm{DF} \in \{1.5, 2, 3\}$ degrees of freedom. For $1 < \mathrm{DF} \leq 2$, the Student's-t distribution leads to residuals with infinite population variance, violating Assumption 1. The plotted empirical coverage remains near the target in these cases. This observation does not extend the validity theorem beyond its assumptions, and the large volumes show a substantial loss of informativeness.



(a) Small sample, $|\mathcal{D}_{\mathrm{cal}}| = 30$. Output dimensions $d = 10$.



(b) Large sample, $|\mathcal{D}_{\mathrm{cal}}| = 500$. Output dimensions $d = 10$.

Figure S17: Performance of the proposed TSCP method and other approaches on simulated data with noise following a Student's-t distribution with varying degrees of freedom, using small (a) and large (b) sample sizes. The target coverage level is 0.9. All methods except *Emp. Copula* have near-nominal coverage in all settings. TSCP has near-nominal empirical coverage in these displayed heavy-tail cases; the infinite-variance cases are outside the stated assumptions. In the small sample setting, Point CHR suffers more in terms of prediction set size due to its inefficient use of calibration data, although it can outperform TSCP in large samples.